Replication: Does Premium Version Adoption in mHealth Improve User Engagement and Health-Related Outcomes?

Jiang, Uetake, and Yang (Marketing Science 2026): the Callaway-Sant'Anna workflow on a freemium app, with causalmetrics

Weiheng Zhang

September 07, 2026

Contents

1 The paper in one page

The question is whether paying for the premium tier of a freemium fitness app changes what users do and what they achieve. Jiang, Uetake, and Yang (2026) observe 11,873 users of a food- and exercise-tracking app for 15 weeks after registration. Some pay $39.99 for a year of premium features in weeks 1 to 7; most never do. The outcomes are four engagement measures (days tracking food, days tracking exercise, days tracking and staying within the calorie budget, exercise calories logged) and two health measures (any weight loss relative to the starting weight, pounds lost).

The answer is an immediate lift that fades and does not reach the scale. Premium adopters track more in the adoption week, the lift attenuates within about five weeks, and weight loss shows at most a modest, delayed, and imprecise improvement. A mechanism section splits adopters by their exposure to the free version before upgrading: the lift is concentrated among low-exposure users, which fits hedonic decline (novelty wears off) rather than sunk costs (everyone paid the same fee).

The methodological point is why this paper anchors Section 07. Adoption is staggered and chosen, and later adopters upgrade while their engagement is already rising. A pooled two-way fixed effects regression therefore overstates both the size and the persistence of the effects. The paper's preferred estimator is Callaway and Sant'Anna (2021) with the doubly robust score, not-yet-treated comparisons, and covariates in both nuisances, followed by the Rambachan and Roth (2023) sensitivity analysis and a matched-sample design for time-varying selection. That is exactly the workflow of the lecture, and every step below runs on the package's functions: `att_gt()`, `aggregate_att()`, `event_study_frame()`, `plot_event_study()`, `honestdid_inputs()`, plus `bacon_decomp()`, `twfe_weights()`, `pretrend_power()`, `did_imputation()`, and `sdid_weights()` in the extensions.

What the replication package allows. The proprietary data are under a nondisclosure agreement. The authors ship two simulated datasets that preserve the panel structure and all variables, the complete analysis code (15 R scripts and one Stata script), and a README that says the scripts are illustrational: numbers from the simulated data differ from the paper. The standard for matching in this note is therefore the authors' scripts run on the simulated data; published numbers are printed for orientation only and labelled as such. Section 3.1 verifies the match standard directly.

2 Data and setting

One row per user-week, with adoption timing, outcomes, and the covariates the design conditions on. `weeks_join` counts weeks since registration (1 to 15). `week_upgrade` is the adoption week; values above 100 mean the user never upgrades in the window. The `first_7_*` columns are week-1 covariates: engagement, weight, and three national Google search indexes for app- and weight-loss-related terms.

```
Data <- read.csv(file.path(data_path, "data", "simulated_data.csv"))
Data$norm_distance <- Data$initial_distance / Data$initialweight
users <- Data %>% arrange(userid, weeks_join) %>% group_by(userid) %>% slice(1) %>% ungroup()
c(users = nrow(users), user_weeks = nrow(Data), weeks = max(Data$weeks_join),
  adopters_w1_7 = sum(users$week_upgrade <= 7), never = sum(users$week_upgrade > 100))
#>         users    user_weeks         weeks adopters_w1_7         never
#>         11873        178095            15           782         11091
```

The estimation sample drops week-1 adopters so every treated user has a pre-period. Adopters in weeks 2 to 7 plus never-adopters form the DiD sample used from Table 3 onward. The group variable `g` codes never-adopters as 0, the package's convention.

```
dat <- Data %>% filter((week_upgrade >= 2 & week_upgrade <= 7) | week_upgrade > 100) %>%
  mutate(g = ifelse(week_upgrade > 100, 0L, week_upgrade),
         rel = ifelse(g == 0, -1000L, weeks_join - week_upgrade),
         rel_b = ifelse(rel <= -4, -1000L, pmin(rel, 13L)))
xv <- c("initialweight", "heightinches", "age", "first_day", "ini_goalweight", "ini_goalbmi",
        "ini_bmi", "gender", "norm_distance",
        "first_7_track_food_avg", "first_7_foodcalories_avg", "first_7_track_exercise_avg",
        "first_7_within_track_avg", "first_7_exercisecalories_avg", "first_7_lower_initial_avg",
        "first_7_weight_loss_norm_avg", "first_7_app_weight_loss_search_avg",
        "first_7_app_premium_search_avg", "first_7_weight_loss_premium_search_avg")
table(users$week_upgrade[users$week_upgrade <= 7])
#>
#>   1   2   3   4   5   6   7
#> 426 118  68  67  40  44  19
```

2.1 Table 1: summary statistics

Table 1 describes the full sample of 11,873 users, per-user averages over the 15 weeks. The published sample is 29 percent male, 42 years old, 204 pounds at the start, aiming to lose 19 percent of the initial weight; food is tracked on 58 percent of days. The simulated data are drawn to preserve these properties.

```
sumdat <- Data %>% filter(week_upgrade <= 7 | week_upgrade > 100) %>%
  arrange(userid, weeks_join) %>% group_by(userid) %>%
  mutate(across(c(track_food_avg, track_exercise_avg, within_track_avg,
                  exercisecalories_avg, foodcalories_avg, budgetcalories_avg), mean,
                .names = "{.col}_mean")) %>% slice(1) %>% ungroup()
t1_vars <- c(gender = "Male", age = "Age", initialweight = "Initial weight (lb)",
             ini_goalweight = "Target weight (lb)", initial_distance = "Initial distance to goal weight (lb)",
             norm_distance = "Proportional distance to goal weight",
             budgetcalories_avg = "Suggested budget of calories (kcal)",
             foodcalories_avg_mean = "Food calories logged (kcal)",
             exercisecalories_avg_mean = "Exercise calories logged (kcal)",
             track_food_avg_mean = "Avg. proportion of days tracked food",
             track_exercise_avg_mean = "Avg. proportion of days tracked exercise",
             within_track_avg_mean = "Avg. proportion of days tracked and within budget",
             active_days = "Number of active days")
t1 <- tibble(Variable = unname(t1_vars),
             Mean = sapply(names(t1_vars), function(v) mean(sumdat[[v]], na.rm = TRUE)),
             `Std. dev.` = sapply(names(t1_vars), function(v) sd(sumdat[[v]], na.rm = TRUE)))
t1 %>% mutate(across(where(is.numeric), ~ fmt(.x))) %>%
  bind_rows(tibble(Variable = "N", Mean = fmt0(nrow(sumdat)), `Std. dev.` = "")) %>%
  kbl(booktabs = TRUE, linesep = "", align = "lrr",
      caption = "Table 1 of the paper: summary statistics for the full sample (simulated data; per-user averages over the 15-wee
  tab_style()
```

2.2 Figure 1: when users upgrade

Adoption is front-loaded and staggered, which is the whole identification problem. Most upgrades happen in week 1; the histogram motivates both the staggered design and the decision to drop week-1 adopters, who have no pre-period.

Table 1: Table 1 of the paper: summary statistics for the full sample (simulated data; per-user averages over the 15-week window).

Variable	Mean	Std. dev.
Male	0.292	0.455
Age	41.717	14.669
Initial weight (lb)	204.070	53.640
Target weight (lb)	162.683	36.724
Initial distance to goal weight (lb)	41.429	34.240
Proportional distance to goal weight	0.186	0.125
Suggested budget of calories (kcal)	1,638.424	432.302
Food calories logged (kcal)	820.017	480.002
Exercise calories logged (kcal)	114.818	152.139
Avg. proportion of days tracked food	0.575	0.250
Avg. proportion of days tracked exercise	0.278	0.238
Avg. proportion of days tracked and within budget	0.483	0.234
Number of active days	165.842	84.635
N	11,873	

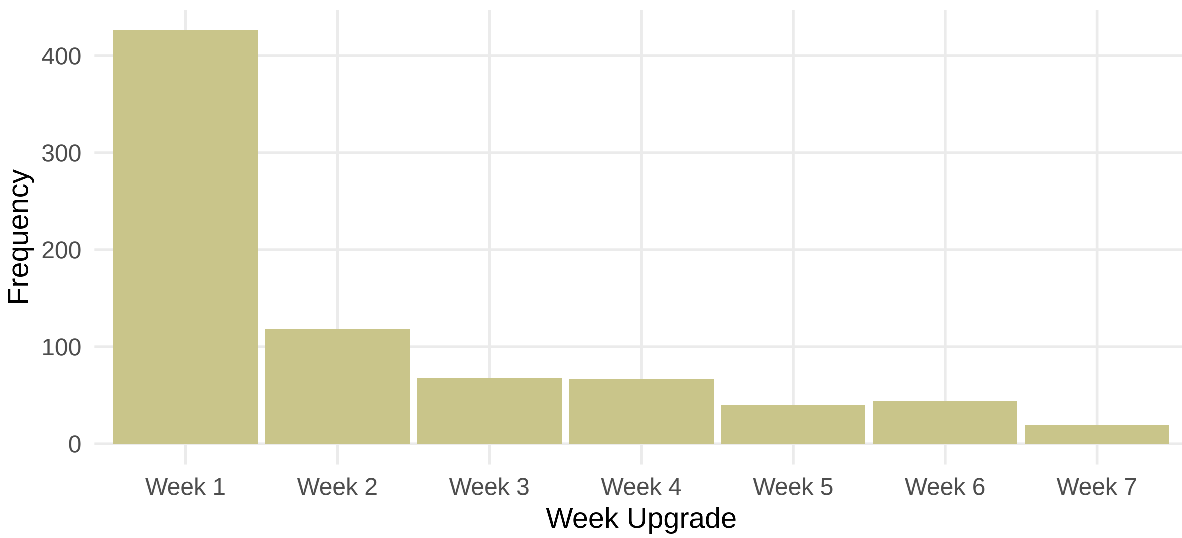


Figure 1: Figure 1 of the paper: number of premium upgrades in each week since registration (simulated data).

2.3 Table 2: who upgrades

Table 2 profiles adopters with a linear probability model and shows why selection is the threat. The published estimates say adopters are older, aim to lose a larger share of their weight, log more exercise calories in week 1, and are less likely to have tracked food consistently in week 1; the app-name premium search index predicts upgrading. The regression uses classical standard errors, as the authors' `lm` call does.

```
f1 <- ever_upgrade ~ initialweight + heightinches + age + first_day + ini_goalweight +
  ini_goalbmi + ini_bmi + norm_distance + gender + first_7_track_food_avg +
  first_7_foodcalories_avg + first_7_track_exercise_avg + first_7_within_track_avg +
  first_7_exercisecalories_avg + first_7_lower_initial_avg + first_7_weight_loss_norm_avg
f2 <- update(f1, . ~ . + first_7_app_premium_search_avg + first_7_app_weight_loss_search_avg +
               first_7_weight_loss_premium_search_avg)
users_r <- users %>% filter((week_upgrade >= 2 & week_upgrade <= 7) | week_upgrade > 100)
m1 <- feols(f1, data = users, vcov = "iid")
m2 <- feols(f2, data = users_r, vcov = "iid")
```

```r
cm <- c(initialweight = "Initial weight", heightinches = "Height", age = "Age",
        first_day = "Start date", ini_goalweight = "Initial goal weight",
        ini_goalbmi = "Initial goal BMI", ini_bmi = "Initial BMI",
        norm_distance = "Prop. distance to goal weight", gender = "Male",
        first_7_track_food_avg = "Prop. days tracked food (wk 1)",
        first_7_foodcalories_avg = "Food calories (wk 1)",
        first_7_track_exercise_avg = "Prop. days tracked exercise (wk 1)",
        first_7_within_track_avg = "Prop. days within budget (wk 1)",
        first_7_exercisecalories_avg = "Exercise calories (wk 1)",
        first_7_lower_initial_avg = "Prop. days below initial weight (wk 1)",
        first_7_weight_loss_norm_avg = "Avg. weight change (wk 1)",
        first_7_app_premium_search_avg = "App premium searches (wk 1)",
        first_7_app_weight_loss_search_avg = "App weight loss searches (wk 1)",
        first_7_weight_loss_premium_search_avg = "Weight loss premium searches (wk 1)")
modelsummary::modelsummary(list("Full sample" = m1, "Restricted sample" = m2),
  output = "kableExtra", coef_map = cm, stars = c(`*` = 0.10, `**` = 0.05, `***` = 0.01),
  gof_map = list(list(raw = "nobs", clean = "Observations", fmt = 0),
                 list(raw = "r.squared", clean = "R2", fmt = 3)),
  fmt = 3, title = "Table 2 of the paper: linear probability model of upgrading within seven weeks (classical standard errors, a
  tab_style("scale_down")
```

3 The staggered DiD design

The estimand is the group-time average treatment effect and everything else is an aggregation of it. For adoption cohort g and week t, $ATT(g,t) = E[Y_t(1) - Y_t(0) \mid G = g]$. Callaway and Sant'Anna (2021) estimate each cell by a doubly robust comparison of the outcome change from $g-1$ to t between cohort g and users not yet treated at t, with the covariates entering a propensity score and an outcome regression. Under limited anticipation and conditional parallel trends, either nuisance being correct suffices. Dynamic aggregation over event time $e = t - g$ gives the event studies of Figures 2 and 3; the simple aggregation gives the overall ATT of Table 3; the multiplier bootstrap gives simultaneous bands.

The specification, exactly as in the authors' scripts.

- Sample: adopters in weeks 2-7 plus never-adopters (11,447 users).
- Comparison group: not yet treated (never-adopters plus later cohorts).
- Method: doubly robust; logistic propensity, linear outcome regression.
- Covariates: demographics, start date, initial goals and BMI, proportional distance to goal, week-1 engagement and weight, week-1 search indexes (19 variables).
- Benchmark: TWFE event study with user and week fixed effects, dummies for event times -3 to 12 and 13+, reference: never-adopters and event times ≤ -4; pooled TWFE with a single post indicator. Standard errors clustered by user everywhere.

3.1 Verifying the match standard

Our `att_gt()` reproduces the `did` package on this data to every digit, and the TWFE helper reproduces the authors' dummy regression exactly. The chunk runs both packages on food tracking and compares; it also compares `fixest::i()` with the authors' hand-built dummies.

```r
a_cm <- att_gt(dat, id = "userid", time = "weeks_join", group = "g", y = "track_food_avg",
               x = xv, method = "dr", control_group = "notyet", n_boot = 0)
s_cm <- aggregate_att(a_cm, type = "simple", n_boot = 0)
b_did <- suppressWarnings(did::att_gt(yname = "track_food_avg", tname = "weeks_join",
  idname = "userid", gname = "week_upgrade", xformla = reformulate(xv), data = dat,
  control_group = "notyettreated", est_method = "dr"))
s_did <- did::aggte(b_did, type = "simple")
tw_i <- feols(track_food_avg ~ i(rel_b, ref = -1000) | userid + weeks_join, data = dat, cluster = ~userid)
da <- dat
for (k in 0:12) da[[paste0("u_", k)]] <- as.integer(da$rel == k)
da$u_13a <- as.integer(da$rel >= 13 & da$g != 0)
for (k in 1:3) da[[paste0("u_p", k)]] <- as.integer(da$rel == -k)
tw_a <- feols(reformulate(c(paste0("u_p", 3:1), paste0("u_", 0:12), "u_13a"),
```

Table 2: Table 2 of the paper: linear probability model of upgrading within seven weeks (classical standard errors, as in the authors' script). The restricted sample is the DiD sample: adopters in weeks 2-7 and never-adopters.

	Full sample	Restricted sample
Initial weight	−0.001	−0.001***
	(0.001)	(0.001)
Height	0.003	0.002
	(0.004)	(0.003)
Age	0.002***	0.001***
	(0.000)	(0.000)
Start date	−0.000	−0.000
	(0.000)	(0.000)
Initial goal weight	0.001	0.002*
	(0.001)	(0.001)
Initial goal BMI	−0.004	−0.009
	(0.007)	(0.005)
Initial BMI	0.006	0.009**
	(0.004)	(0.003)
Prop. distance to goal weight	0.110**	0.048
	(0.043)	(0.034)
Male	−0.009	−0.015**
	(0.008)	(0.006)
Prop. days tracked food (wk 1)	−0.052***	−0.047***
	(0.019)	(0.015)
Food calories (wk 1)	0.000	0.000
	(0.000)	(0.000)
Prop. days tracked exercise (wk 1)	−0.014	−0.006
	(0.010)	(0.008)
Prop. days within budget (wk 1)	−0.014	−0.011
	(0.013)	(0.010)
Exercise calories (wk 1)	0.000***	0.000
	(0.000)	(0.000)
Prop. days below initial weight (wk 1)	0.030***	0.007
	(0.008)	(0.006)
Avg. weight change (wk 1)	−0.002	−0.001
	(0.001)	(0.001)
App premium searches (wk 1)		0.037***
		(0.010)
App weight loss searches (wk 1)		−0.031
		(0.026)
Weight loss premium searches (wk 1)		−0.016*
		(0.008)
Observations	11 873	11 447
R2	0.016	0.014

* p < 0.1, ** p < 0.05, *** p < 0.01

```
                              response = "track_food_avg"), data = da,
               fixef = c("userid", "weeks_join"), cluster = ~userid)
c(att_gt_vs_did = abs(s_cm$overall$estimate - s_did$overall.att),
  twfe_i_vs_dummies = max(abs(sort(coef(tw_i)) - sort(coef(tw_a)))))
#>     att_gt_vs_did twfe_i_vs_dummies
#>      2.280606e-12      0.000000e+00
```

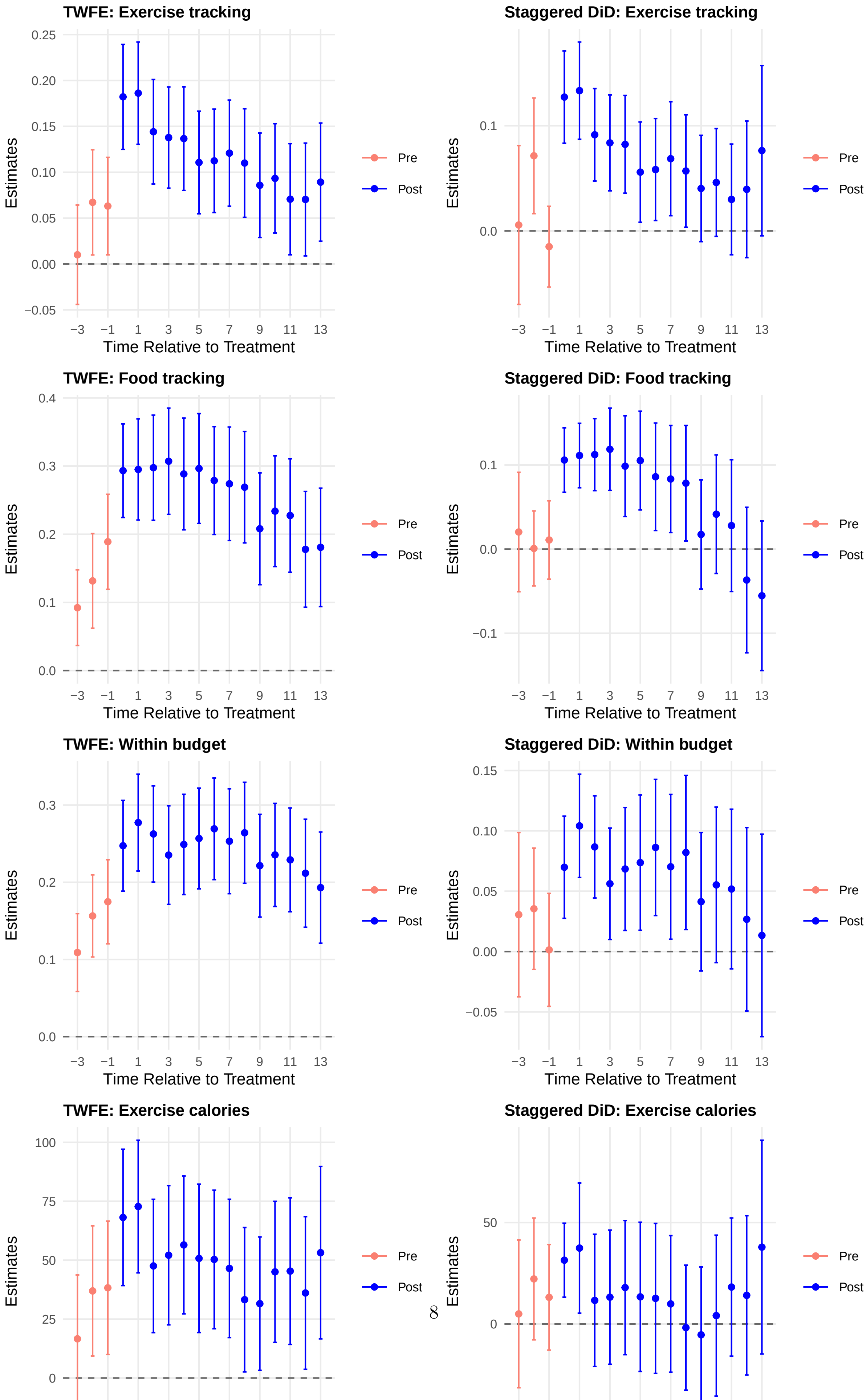
Both differences are zero to machine precision. From here on, every Callaway-Sant'Anna object is the package's, and every published number is quoted for orientation.

4 Results: event studies and the overall ATT

```
cs_main <- lapply(setNames(outcomes6, outcomes6), function(y)
  att_gt(dat, id = "userid", time = "weeks_join", group = "g", y = y, x = xv,
         method = "dr", control_group = "notyet", n_boot = n_boot, seed = 678))
dyn_main <- lapply(cs_main, aggregate_att, type = "dynamic", min_e = -3, seed = 678)
simple_main <- lapply(cs_main, aggregate_att, type = "simple", seed = 678)
tw_main <- lapply(setNames(outcomes6, outcomes6), function(y) tw_es(y, dat))
```

4.1 Figures 2 and 3: TWFE against Callaway-Sant'Anna

The two columns of each figure are the paper's argument in one picture. The left column is the TWFE event study: a sharp rise at adoption that appears to persist for three months. The right column is the Callaway-Sant'Anna event study with simultaneous bands: the same sharp rise, but it attenuates within weeks. The published reading is that TWFE overstates persistence because later adopters upgrade on rising trajectories and contaminate the comparisons.


TWFE: Exercise tracking
Staggered DiD: Exercise tracking
TWFE: Food tracking
Staggered DiD: Food tracking
TWFE: Within budget
Staggered DiD: Within budget
TWFE: Exercise calories
Staggered DiD: Exercise calories
Estimates
Time Relative to Treatment
Pre
Post

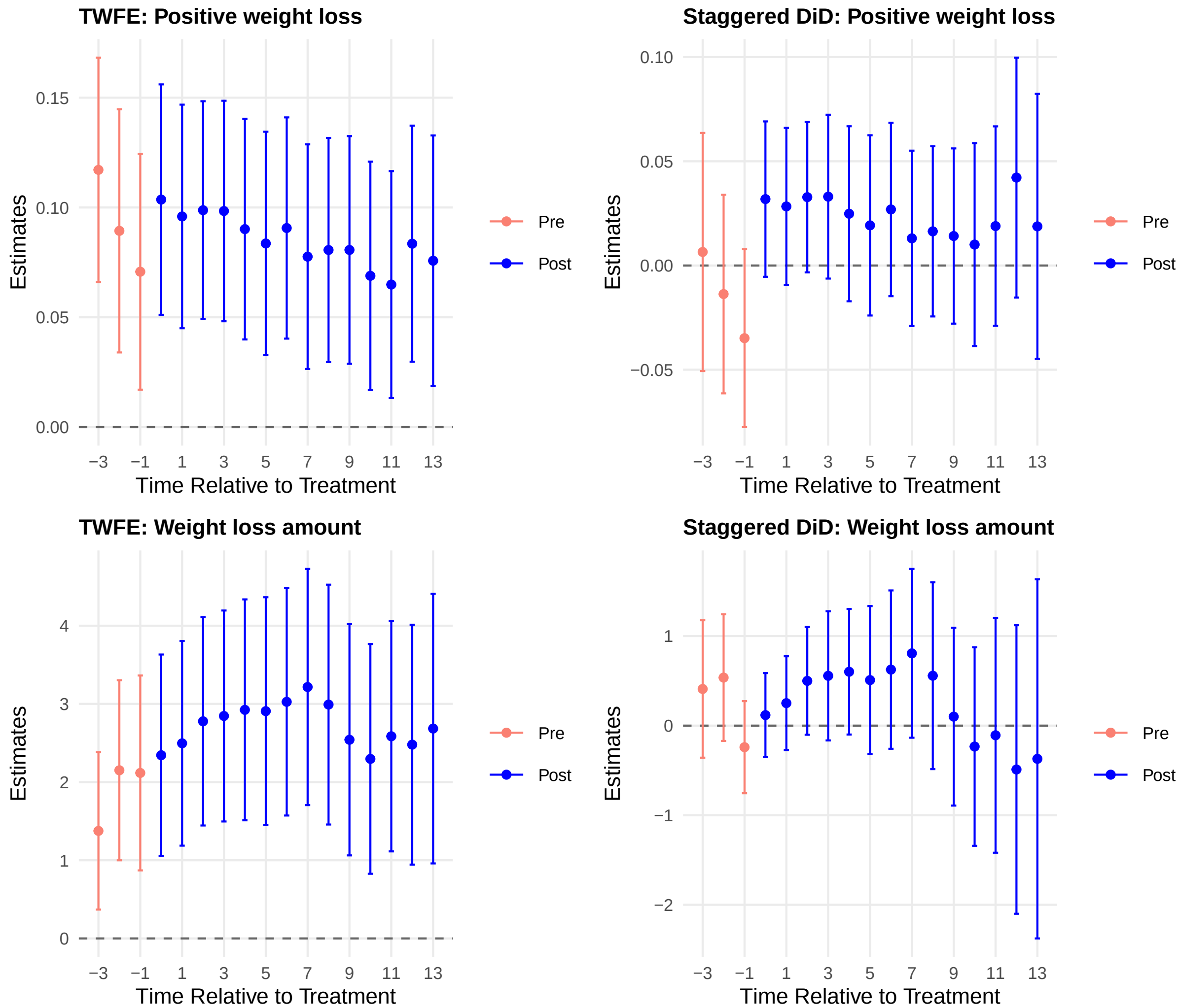


Figure 3: Figure 3 of the paper: impact of premium adoption on the two health outcomes, same layout as Figure 2.

On the simulated data the levels and the TWFE-CS ordering reproduce; the decay is clear for the exercise outcomes and slower for the food outcomes. Exercise tracking falls from 0.127 in the adoption week to 0.056 five weeks later, and exercise calories from 31.5 to 13.4. Food tracking starts at 0.106, is still 0.105 at week five, and declines to 0.041 by week ten; budget adherence behaves like food tracking. The published estimates attenuate faster (food tracking insignificant after week five), so this simulated draw damps the decay without changing its direction. The health outcomes show wide bands and no effect covered by the band at any horizon.

4.2 Table 3: overall ATT, pooled TWFE against Callaway-Sant'Anna

The overall ATT makes the overstatement quantitative. The pooled TWFE coefficient (one post indicator, user and week fixed effects) is compared with the simple aggregation of the group-time effects.

```
tw_pool <- lapply(setNames(outcomes6, outcomes6), function(y)
  feols(as.formula(paste0(y, " ~ upgraded | userid + weeks_join")), data = dat, cluster = ~userid))
row_tw <- function(y) {
  ct <- coeftable(tw_pool[[y]])["upgraded", ]
  tibble(Variable = labels6[y], ATT = fmt(ct[1]), `Std. error` = fmt(ct[2]), `95\\% CI` = ci_of(ct[1], ct[2]))
}
row_cs <- function(y) {
```

```r
  o <- simple_main[[y]]$overall
  tibble(Variable = labels6[y], ATT = fmt(o$estimate), `Std. error` = fmt(o$std.error), `95\\% CI` = ci_of(o$estimate, o$std.err
}
pub3 <- tibble(Variable = labels6,
  `Published TWFE` = c("0.124", "0.163", "0.124", "47.657", "0.036", "1.584"),
  `Published CS` = c("0.074", "0.060", "0.048", "32.183", "0.026", "0.487"))
bind_rows(bind_rows(lapply(outcomes6, row_tw)), bind_rows(lapply(outcomes6, row_cs))) %>%
  left_join(pub3, by = "Variable") %>%
  mutate(`Published ATT` = c(pub3$`Published TWFE`, pub3$`Published CS`)) %>%
  select(Variable, ATT, `Std. error`, `95\\% CI`, `Published ATT`) %>%
  kbl(booktabs = TRUE, linesep = "", escape = FALSE, align = "lcccc",
      caption = "Table 3 of the paper: overall ATT estimates from the pooled TWFE regression and from the simple aggregation of
  pack_rows("Panel A: pooled TWFE estimates", 1, 6) %>%
  pack_rows("Panel B: Callaway and Sant'Anna (2021) estimates", 7, 12) %>%
  tab_style()
```

Table 3: Table 3 of the paper: overall ATT estimates from the pooled TWFE regression and from the simple aggregation of the Callaway-Sant'Anna group-time effects, standard errors clustered by user. The published column shows the proprietary-data estimates for orientation.

Variable	ATT	Std. error	95% CI	Published ATT
Panel A: pooled TWFE estimates				
Exercise tracking	0.083	0.015	[0.053, 0.112]	0.124
Food tracking	0.147	0.019	[0.110, 0.185]	0.163
Within budget	0.121	0.017	[0.088, 0.154]	0.124
Exercise calories	22.644	8.402	[6.177, 39.111]	47.657
Positive weight loss	0.016	0.015	[-0.013, 0.046]	0.036
Weight loss amount	1.116	0.333	[0.463, 1.768]	1.584
Panel B: Callaway and Sant'Anna (2021) estimates				
Exercise tracking	0.073	0.013	[0.048, 0.098]	0.074
Food tracking	0.076	0.016	[0.044, 0.107]	0.060
Within budget	0.068	0.014	[0.040, 0.095]	0.048
Exercise calories	14.344	9.140	[-3.570, 32.259]	32.183
Positive weight loss	0.023	0.013	[-0.002, 0.049]	0.026
Weight loss amount	0.322	0.248	[-0.164, 0.808]	0.487

TWFE is larger for the four engagement outcomes and the weight-loss amount, here by factors of 1.1 to 3.5. Positive weight loss is small and imprecise in both panels, and its two estimates are indistinguishable. In the paper the ratios run from about 1.5 (exercise calories) to 3.3 (weight loss amount). Extension 8.1 decomposes where the gap comes from.

5 Sensitivity and robustness

5.1 Figure 4: overlap

The overlap assumption is checked by the propensity-score densities by cohort. The score is a pooled logistic regression of the premium indicator on the design covariates plus first-week weight, as in the authors' script; nondegenerate, overlapping densities support the weighting.

```r
ps_x <- c(xv, "first_7_weight_avg")
ps_fit <- glm(reformulate(ps_x, response = "upgraded"), data = dat, family = binomial())
dat$pscore <- predict(ps_fit, newdata = dat, type = "response")
dat$cohort_lab <- ifelse(dat$g == 0, "Never", paste0("Week ", dat$g))
ggplot(dat, aes(x = pscore, colour = cohort_lab)) +
  geom_density() +
  scale_colour_manual(values = c("Week 2" = "orange", "Week 3" = "blue", "Week 4" = "brown",
                                 "Week 5" = "green", "Week 6" = "pink", "Week 7" = "darkgreen",
                                 "Never" = "purple"), name = "Upgrade Week") +
```

```
  coord_cartesian(xlim = c(0, 0.1), ylim = c(0, 60)) +
  labs(x = "Predicted Propensity Score", y = "Density") +
  theme_classic(base_size = 11) + theme(legend.position = "top")
```

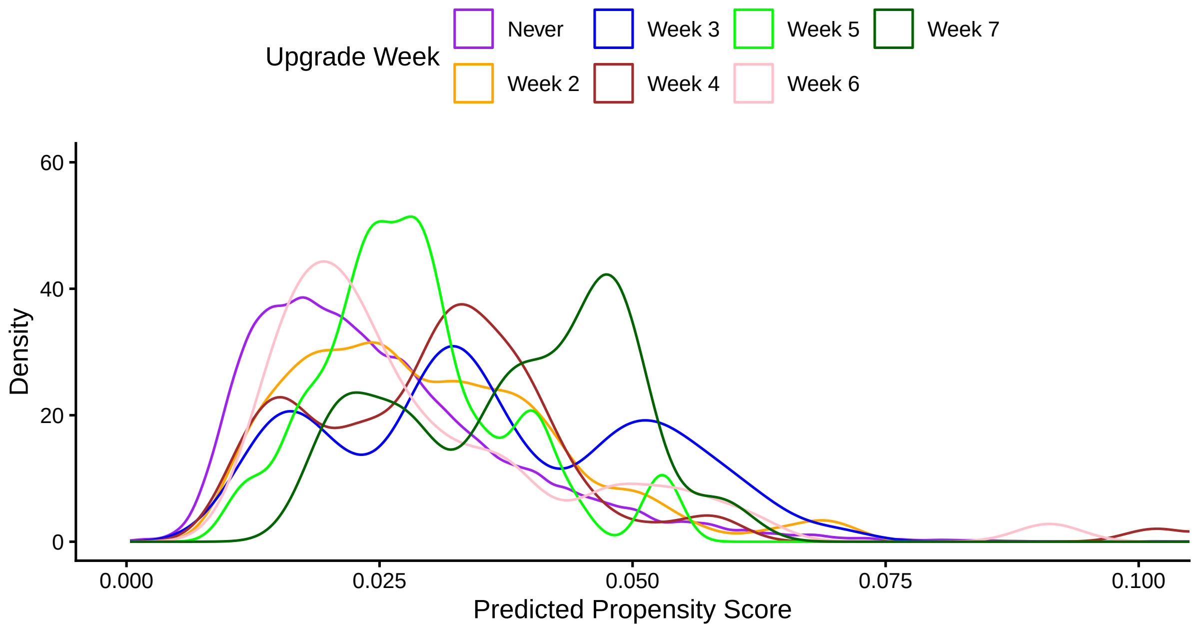


Figure 4: Figure 4 of the paper: density of the predicted propensity score across adoption cohorts (pooled logistic regression on the design covariates plus first-week weight).

5.2 Figures 5 and 6: pre-adoption trajectories

The raw trajectories are the selection story. Mean outcomes by week since registration, one line per adoption cohort (including week-1 adopters) plus never-adopters, with each cohort's adoption week marked. In the paper, later adopters ride rising engagement into their upgrade and never-adopters drift down; weight-loss trajectories rise for everyone, steepest for late adopters.

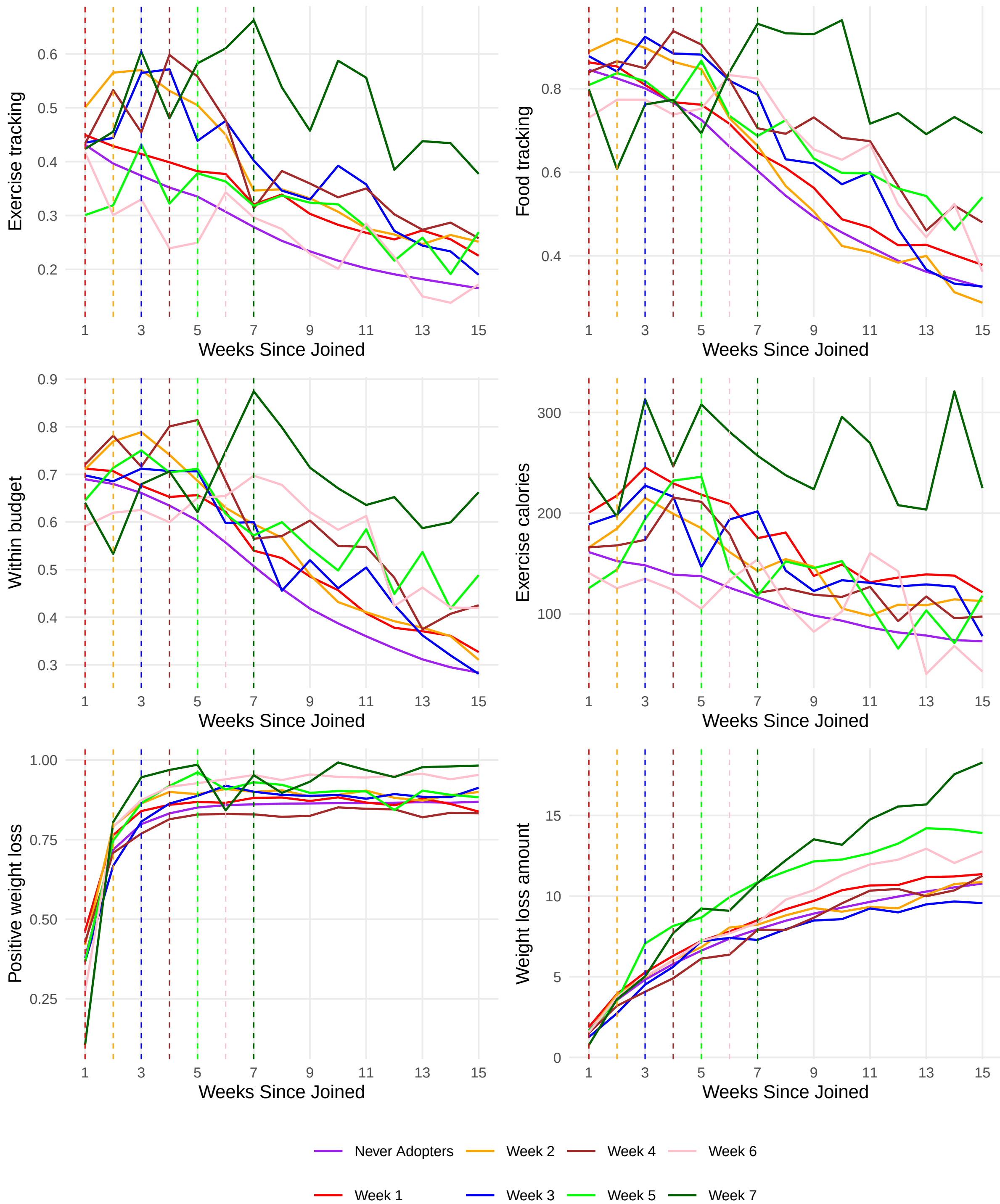


Figure 5: Figures 5 and 6 of the paper: pre-adoption trajectories. Mean outcome by week since registration for each adoption cohort and for never-adopters; dashed vertical lines mark adoption weeks.

5.3 Figure 7: Rambachan-Roth sensitivity

The sensitivity analysis asks how large a violation of conditional parallel trends the significant effects survive. The relative-magnitudes family bounds the post-treatment violation by $\bar{M}$ times the largest pre-treatment violation; the conditional least-favorable hybrid interval is computed for the week-of-adoption effect over a grid of $\bar{M}$. `att_gt()` is re-run with the universal base period, so every pre-period coefficient is a genuine pre-trend estimate, and `honestdid_inputs()` hands the coefficients and their clustered covariance to HonestDiD. The published thresholds are $\bar{M} \leq 1.6$ for exercise tracking, ≤ 1.1 for food tracking, and ≤ 1.0 for the other two engagement outcomes.

```
cs_uni <- lapply(setNames(eng4, eng4), function(y)
  att_gt(dat, id = "userid", time = "weeks_join", group = "g", y = y, x = xv,
         method = "dr", control_group = "notyet", base_period = "universal", n_boot = 0))
dyn_uni <- lapply(cs_uni, aggregate_att, type = "dynamic", min_e = -4, max_e = 3, n_boot = 0)
```

```
honest_one <- function(dyn, mbars = seq(0.5, 1.8, by = 0.1)) {
  hi <- honestdid_inputs(dyn)
  l1 <- HonestDiD::basisVector(1, hi$numPostPeriods)
  orig <- HonestDiD::constructOriginalCS(betahat = hi$betahat, sigma = hi$sigma,
    numPrePeriods = hi$numPrePeriods, numPostPeriods = hi$numPostPeriods, l_vec = l1)
  rob <- HonestDiD::createSensitivityResults_relativeMagnitudes(betahat = hi$betahat,
    sigma = hi$sigma, numPrePeriods = hi$numPrePeriods, numPostPeriods = hi$numPostPeriods,
    Mbarvec = mbars, l_vec = l1, gridPoints = 100)
  list(orig = as.data.frame(orig), rob = as.data.frame(rob))
}
honest_res <- lapply(dyn_uni, honest_one)
```

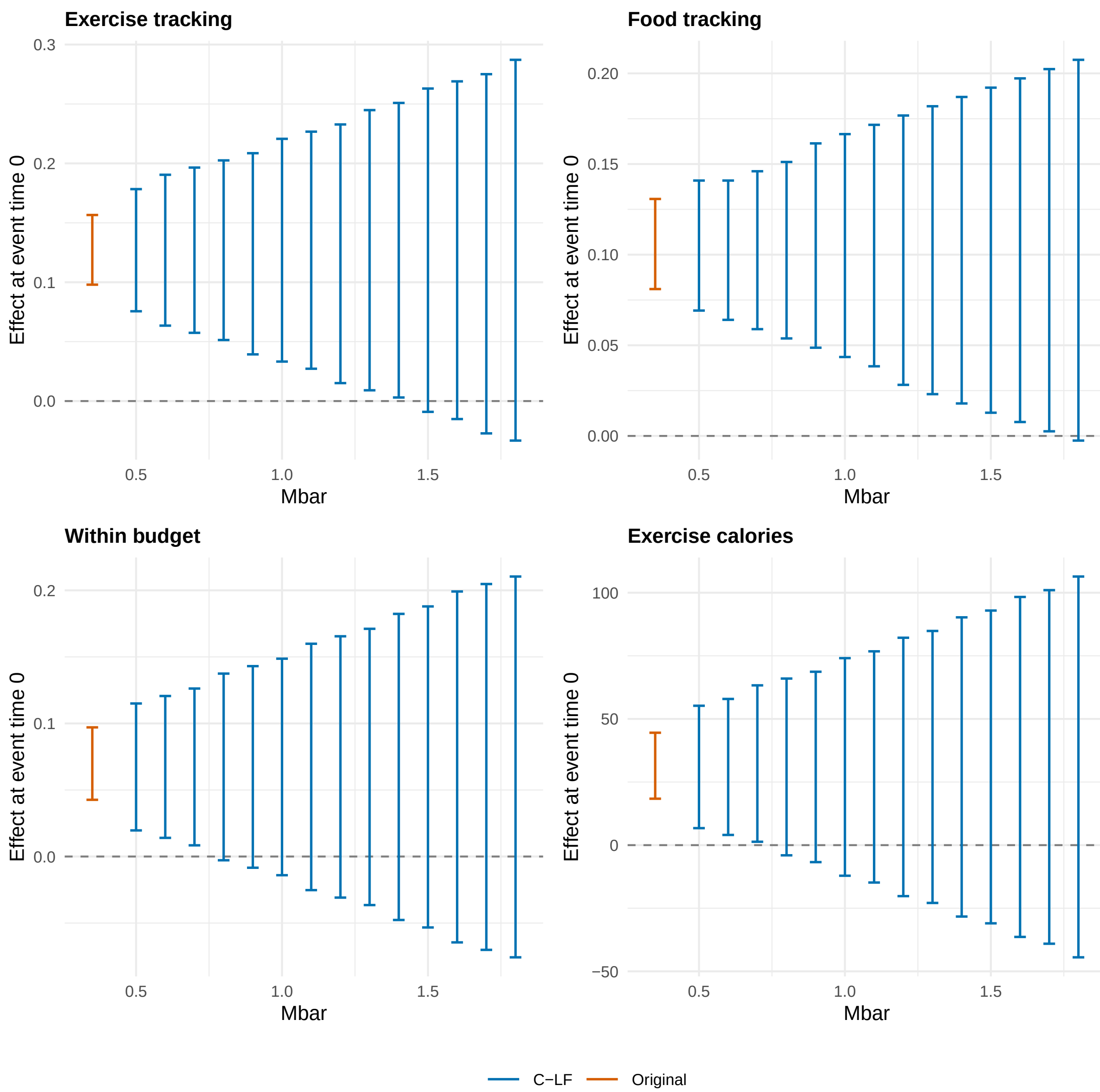


Figure 6: Figure 7 of the paper: conditional least-favorable hybrid confidence intervals for the week-of-adoption effect under relative-magnitudes violations of size Mbar times the largest pre-treatment violation. The leftmost red interval is the original confidence interval under conditional parallel trends.

On the simulated data the week-of-adoption effects stay significant up to Exercise tracking: 1.4; Food tracking: 1.7; Within budget: 0.7; Exercise calories: 0.7. The reading is the paper's: the effects survive violations comparable to the worst pre-period deviation, and extreme violations erase them. Extension 8.4 extends the exercise to more horizons and the smoothness family, on a finer grid.

5.4 Figure 8: matched sample for time-varying selection

The matched design conditions on recent trajectories, not only week-1 levels. Adopters in weeks 5 and 6 are matched one-to-ten to never-adopters on demographics, goals, week-1 engagement, and weeks 3-4 engagement, weight, and search intensities, with a generalized additive model propensity (the authors' seed 123). A TWFE event study on the matched panel then adjusts for what the matching balanced.

```r
recent <- Data %>% arrange(userid, weeks_join) %>% group_by(userid) %>%
  mutate(across(c(track_food_avg, foodcalories_avg, track_exercise_avg, within_track_avg,
                  exercisecalories_avg, lower_initial_avg, weight_loss_norm_avg, weight_avg,
                  app_premium_search_avg, app_weight_loss_search_avg, weight_loss_premium_search_avg),
                list(w3 = ~nth(.x, 3), w4 = ~nth(.x, 4), w5 = ~nth(.x, 5)),
                .names = "{.col}_{.fn}")) %>% ungroup()
sub8 <- recent %>% group_by(userid) %>%
  filter(week_upgrade %in% c(5, 6) | ever_upgrade == 0) %>% slice(1) %>% ungroup()
f8 <- reformulate(c("initialweight", "heightinches", "age", "first_day", "ini_goalweight",
  "ini_goalbmi", "ini_bmi", "gender", "norm_distance",
  "first_7_track_food_avg", "first_7_foodcalories_avg", "first_7_track_exercise_avg",
  "first_7_within_track_avg", "first_7_exercisecalories_avg",
  paste0(c("track_food_avg", "foodcalories_avg", "track_exercise_avg", "within_track_avg",
           "exercisecalories_avg", "lower_initial_avg", "weight_loss_norm_avg", "weight_avg",
           "app_weight_loss_search_avg", "app_premium_search_avg", "weight_loss_premium_search_avg"),
         rep(c("_w3", "_w4"), each = 11))), response = "ever_upgrade")
set.seed(123)
m8 <- matchit(f8, data = sub8, method = "nearest", distance = "gam", ratio = 10, replacement = FALSE)
ids8 <- match.data(m8)$userid
dat8 <- Data %>% filter(userid %in% ids8) %>%
  mutate(g = ifelse(week_upgrade > 100, 0L, week_upgrade),
         rel = ifelse(g == 0, -1000L, weeks_join - week_upgrade),
         rel_b = ifelse(rel <= -4, -1000L, pmin(rel, 13L)))
c(adopters = sum(sub8$ever_upgrade), matched_users = length(ids8))
#>      adopters matched_users
#>            84           924
```

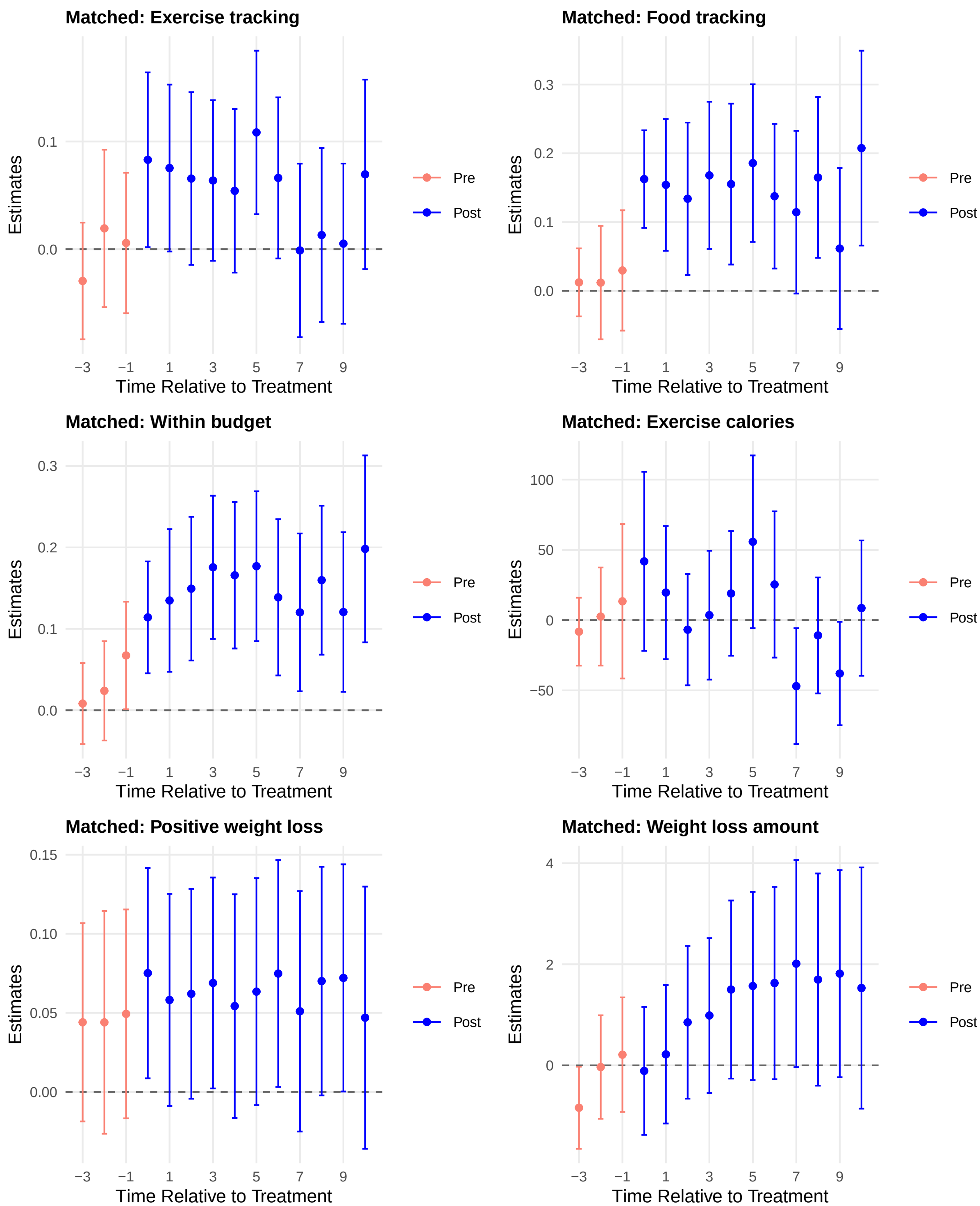


Figure 7: Figure 8 of the paper: TWFE event studies on the matched sample (adopters in weeks 5-6 matched 1:10 to never-adopters on demographics, goals, week-1 engagement, and weeks 3-4 engagement, weight, and search intensities). Pointwise 95 percent intervals, clustered by user.

The immediate lift and the attenuation survive the matching, and the pre-periods flatten. That is the published conclusion: the response is not only a motivational surge that jointly causes upgrading and engagement. Extension 8.5 re-estimates this design with the main estimator instead of TWFE.

6 Mechanisms: exposure to the free version

6.1 Tables 4 and 5: the exposure split

Hedonic decline and sunk costs both predict the immediate lift; only hedonic decline predicts that prior exposure dampens it. Adopters are split at the median days of tracking before the upgrade. Table 4 shows the two groups are otherwise comparable. Table 5 estimates the Callaway-Sant'Anna effects separately for each group against the never-adopters, at the adoption week and one week after.

```
med_track <- median(users$total_track_before[users$week_upgrade >= 2 & users$week_upgrade <= 7])
never_ids <- users$userid[users$week_upgrade > 100]
dat_exp <- dat %>% mutate(distance_goal = weight_avg - goalweight)
adopt_at <- dat_exp %>% filter(weeks_join == week_upgrade)
low_at <- adopt_at %>% filter(total_track_before < med_track)
high_at <- adopt_at %>% filter(total_track_before >= med_track)
t4_vars <- c(total_track_before = "Number of prior days tracked food/exercise", gender = "Male",
             age = "Age", initialweight = "Initial weight (lb)", heightinches = "Height (inch)",
             ini_goalweight = "Initial goal weight (lb)", ini_goalbmi = "Initial goal BMI",
             weight_avg = "Current weight (lb)", distance_goal = "Current distance to goal (lb)")
t4 <- bind_rows(lapply(names(t4_vars), function(v) {
  tt <- t.test(low_at[[v]], high_at[[v]])
  tibble(Variable = t4_vars[v], `Low exposure` = fmt(mean(low_at[[v]], na.rm = TRUE)),
         `High exposure` = fmt(mean(high_at[[v]], na.rm = TRUE)),
         `t-statistic` = fmt(tt$statistic), `p-value` = fmt(tt$p.value))
}))
t4 %>% bind_rows(tibble(Variable = "Number of observations", `Low exposure` = fmt0(nrow(low_at)),
                        `High exposure` = fmt0(nrow(high_at)), `t-statistic` = "", `p-value` = "")) %>%
  kbl(booktabs = TRUE, linesep = "",
      caption = "Table 4 of the paper: comparison of means between low- and high-exposure adopters (median split on days of food
  tab_style()
```

Table 4: Table 4 of the paper: comparison of means between low- and high-exposure adopters (median split on days of food or exercise tracking before the upgrade), with two-sample t-tests.

Variable	Low exposure	High exposure	t-statistic	p-value
Number of prior days tracked food/exercise	8.006	26.695	-27.429	0.000
Male	0.260	0.299	-0.820	0.413
Age	46.148	47.075	-0.622	0.534
Initial weight (lb)	212.764	211.594	0.244	0.807
Height (inch)	66.538	66.586	-0.124	0.901
Initial goal weight (lb)	167.387	165.806	0.422	0.673
Initial goal BMI	26.427	26.142	0.610	0.542
Current weight (lb)	209.934	204.820	1.050	0.294
Current distance to goal (lb)	43.627	36.900	2.062	0.040
Number of observations	169	187		

```
dat_low <- dat %>% filter((total_track_before < med_track & g > 0) | userid %in% never_ids)
dat_high <- dat %>% filter((total_track_before >= med_track & g > 0) | userid %in% never_ids)
t5_run <- function(data) lapply(setNames(eng4, eng4), function(y) {
  a <- att_gt(data, id = "userid", time = "weeks_join", group = "g", y = y, x = xv,
              method = "dr", control_group = "notyet", n_boot = 499, seed = 1234)
  aggregate_att(a, type = "dynamic", min_e = -3, na.rm = TRUE, seed = 1234)$by
})
t5_low <- t5_run(dat_low)
t5_high <- t5_run(dat_high)
```

Low-exposure adopters respond more, as the paper reports. Food tracking one week after upgrading rises by 0.120 for low-exposure adopters and 0.093 for high-exposure adopters (published: 0.167 against 0.041, a wider gap than this simulated draw produces). The same ordering holds for every engagement outcome, most sharply for exercise calories. The sunk cost is the same $39.99 for everyone, so the asymmetry points to hedonic decline.

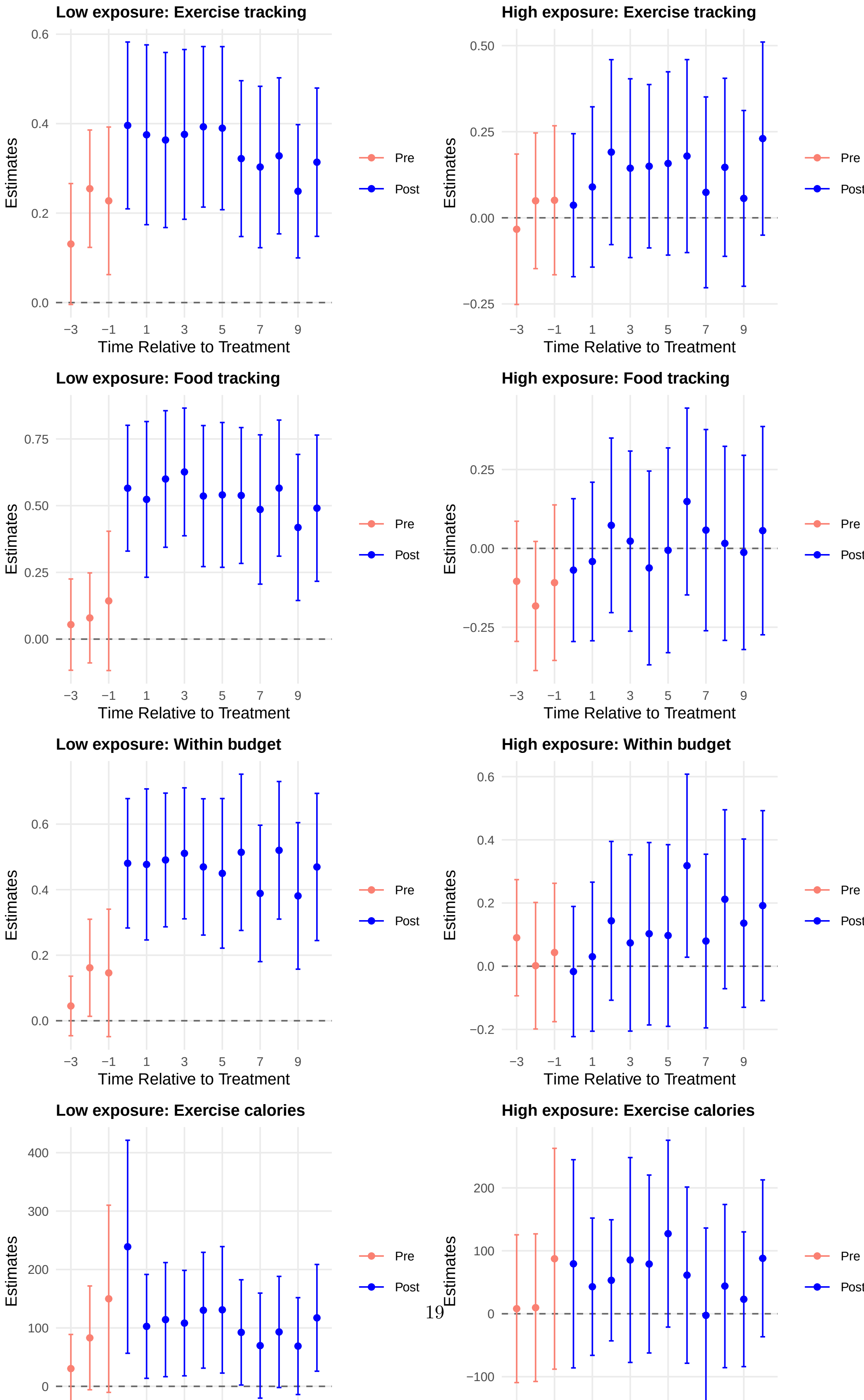
Table 5: Table 5 of the paper: heterogeneous effects by pre-adoption exposure. Callaway-Sant'Anna dynamic effects at the adoption week and one week after, estimated separately for low- and high-exposure adopters against never-adopters.

	Immediate (week 0)		Short term (week 1)	
Outcome	Estimate	Std. error	Estimate	Std. error
Exercise tracking (low exposure)	0.181	0.027	0.204	0.026
Exercise tracking (high exposure)	0.078	0.018	0.069	0.019
Food tracking (low exposure)	0.116	0.017	0.120	0.019
Food tracking (high exposure)	0.084	0.018	0.093	0.020
Within budget (low exposure)	0.106	0.020	0.144	0.020
Within budget (high exposure)	0.031	0.019	0.061	0.020
Exercise calories (low exposure)	48.130	11.482	77.557	13.843
Exercise calories (high exposure)	18.666	8.462	2.698	17.091

6.2 Figures 9 and 10: matched samples by exposure

The sharpest version matches each exposure group to its own controls. Week-5 adopters are split at their own median, each group is matched to never-adopters on recent (weeks 3-4) engagement and search (the authors' probit, ratio 2, seed 123 for the low group; GAM, ratio 1, seed 234 for the high group), and TWFE event studies run per group.

```
med5 <- median(users$total_track_before[users$week_upgrade == 5])
f910 <- reformulate(c("initialweight", "heightinches", "age", "first_day", "ini_goalweight",
  "ini_goalbmi", "ini_bmi", "gender",
  "track_food_avg_w3", "track_exercise_avg_w3",
  paste0(c("track_food_avg", "foodcalories_avg", "track_exercise_avg", "within_track_avg",
           "exercisecalories_avg", "lower_initial_avg", "weight_loss_norm_avg",
           "app_weight_loss_search_avg", "app_premium_search_avg", "weight_loss_premium_search_avg"),
         "_w4")), response = "ever_upgrade")
match_ids <- function(cut_low, seed, dist, ratio, link = NULL) {
  sub <- recent %>% group_by(userid) %>%
    filter(week_upgrade == 5 | ever_upgrade == 0) %>% slice(1) %>% ungroup() %>%
    filter(if (cut_low) total_track_before < med5 else total_track_before >= med5)
  set.seed(seed)
  args <- list(f910, data = sub, method = "nearest", distance = dist, ratio = ratio, replacement = FALSE)
  if (!is.null(link)) args$link <- link
  match.data(do.call(matchit, args), data = sub)$userid
}
ids_low5 <- match_ids(TRUE, 123, "glm", 2, link = "probit")
ids_high5 <- match_ids(FALSE, 234, "gam", 1)
panel_of <- function(ids) Data %>% filter(userid %in% ids) %>%
  mutate(g = ifelse(week_upgrade > 100, 0L, week_upgrade),
         rel = ifelse(g == 0, -1000L, weeks_join - week_upgrade),
         rel_b = ifelse(rel <= -4, -1000L, pmin(rel, 13L)))
pan_low5 <- panel_of(ids_low5); pan_high5 <- panel_of(ids_high5)
c(low = length(ids_low5), high = length(ids_high5))
#>  low high
#>   60   40
```


Low exposure: Exercise tracking
High exposure: Exercise tracking
Low exposure: Food tracking
High exposure: Food tracking
Low exposure: Within budget
High exposure: Within budget
Low exposure: Exercise calories
High exposure: Exercise calories
Estimates
Time Relative to Treatment
Pre
Post

Only the low-exposure group shows the clear lift, as published. The high-exposure panels sit near zero throughout. The same design for week-6 adopters (published as Online Figures D1 and D2) is reproduced in Section 7.5.

7 Appendix replications

7.1 Appendix A: alternative sample

The published check re-runs everything on weeks 2-6 adopters and a 3 percent random sample of never-adopters. The point is that the baseline pattern is not driven by the week-7 cohort (19 users here) or by the size of the control pool.

```
set.seed(123)
never_3pct <- sample(never_ids, round(0.03 * length(never_ids)))
dat_a <- dat %>% filter((g >= 2 & g <= 6) | userid %in% never_3pct)
cs_a <- lapply(setNames(outcomes6, outcomes6), function(y)
  att_gt(dat_a, id = "userid", time = "weeks_join", group = "g", y = y, x = xv,
         method = "dr", control_group = "notyet", n_boot = 499, seed = 678))
dyn_a <- lapply(cs_a, aggregate_att, type = "dynamic", min_e = -3, seed = 678)
```

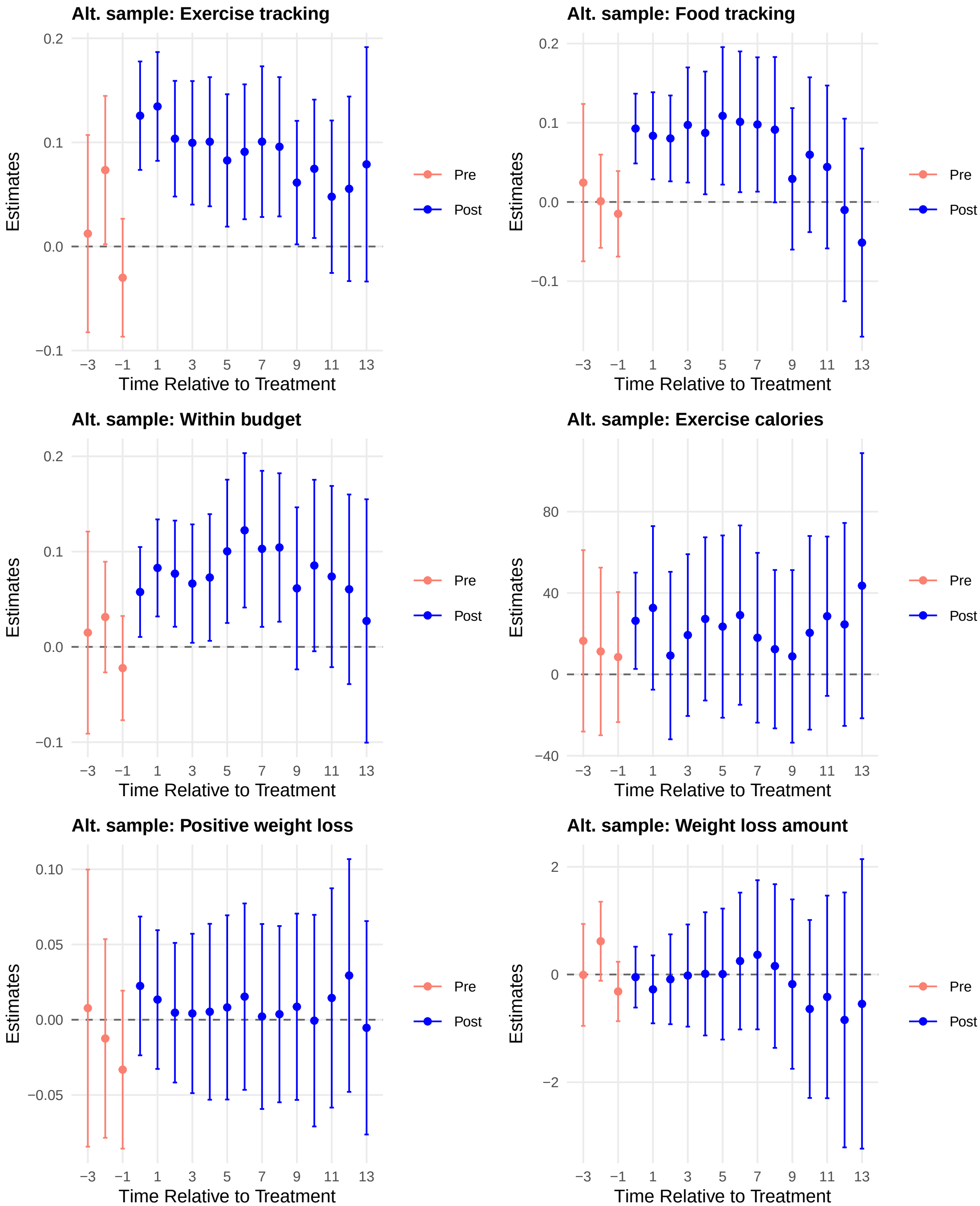


Figure 9: Appendix Figures A1-A2 of the paper (Callaway-Sant'Anna panels): event studies on the alternative sample of weeks 2-6 adopters and a 3 percent random sample of never-adopters.

7.2 Appendix B1: measurement-thresholded outcomes

The second simulated file applies physiologically credible thresholds to food and weight logs. The Callaway-Sant'Anna overall ATTs on that file sit next to the baseline Table 3 panel.

```r
DataB <- read.csv(file.path(data_path, "data", "simulated_data_robust_accuracy.csv"))
DataB$norm_distance <- DataB$initial_distance / DataB$initialweight
dat_b <- DataB %>% filter((week_upgrade >= 2 & week_upgrade <= 7) | week_upgrade > 100) %>%
  mutate(g = ifelse(week_upgrade > 100, 0L, week_upgrade))
simple_b <- lapply(setNames(outcomes6, outcomes6), function(y) {
  a <- att_gt(dat_b, id = "userid", time = "weeks_join", group = "g", y = y, x = xv,
              method = "dr", control_group = "notyet", n_boot = 499, seed = 678)
  aggregate_att(a, type = "simple", seed = 678)$overall
})
```

Table 6: Appendix Table B1 of the paper: Callaway-Sant'Anna overall ATTs on the accuracy-thresholded data next to the baseline estimates (standard errors in parentheses).

Variable	Baseline ATT	Thresholded ATT
Exercise tracking	0.073 (0.013)	0.073 (0.013)
Food tracking	0.076 (0.016)	0.066 (0.015)
Within budget	0.068 (0.014)	0.068 (0.014)
Exercise calories	14.344 (9.140)	14.512 (9.320)
Positive weight loss	0.023 (0.013)	0.024 (0.013)
Weight loss amount	0.322 (0.248)	0.353 (0.246)

7.3 Appendix C1: de Chaisemartin and d'Haultfoeuille

The first alternative estimator replaces the Callaway-Sant'Anna aggregation with the de Chaisemartin-d'Haultfoeuille dynamic estimator. The authors run `did_multiplegt_dyn` with 13 effects and 2 placebos, clustered by user; the table mirrors their Panel B (the average total effect per switcher).

```r
suppressPackageStartupMessages(library(polars))
dcdh <- lapply(setNames(outcomes6, outcomes6), function(y) {
  pdf(NULL)
  m <- DIDmultiplegtDYN::did_multiplegt_dyn(as.data.frame(dat), y, "userid", "weeks_join",
                                            "upgraded", effects = 13, placebo = 2,
                                            cluster = "userid", graph_off = TRUE)
  dev.off()
  m$results$ATE
})
```

Table 7: Appendix Table C1 of the paper: de Chaisemartin-d'Haultfoeuille average effect per switcher (13 effects, 2 placebos, clustered by user), with the Callaway-Sant'Anna simple ATT for comparison.

Variable	ATT	Std. error	95% CI	CS ATT
Exercise tracking	0.061	0.016	[0.029, 0.092]	0.073
Food tracking	0.077	0.016	[0.045, 0.109]	0.076
Within budget	0.073	0.015	[0.043, 0.103]	0.068
Exercise calories	11.942	10.288	[-8.222, 32.106]	14.344
Positive weight loss	0.016	0.016	[-0.016, 0.048]	0.023
Weight loss amount	0.576	0.256	[0.073, 1.079]	0.322

The two heterogeneity-robust estimators agree on the engagement outcomes, as the paper reports for the proprietary data. The weight-loss amount differs more, with both estimates inside each other's wide intervals. Both estimators sit well below the pooled TWFE of Table 3 for engagement.

7.4 Appendix C2: synthetic DiD

The second alternative is synthetic DiD, which the authors run in Stata cohort by cohort. `sdid_weights()` is the package's implementation; it requires a block design, so it is applied per adoption cohort and the cohort estimates are combined with weights proportional to the number of treated cells, which is how the Stata command aggregates. Cohort 2 has a single pre-period and is excluded (the regularization needs two); no Stata output ships with the package, so these numbers stand without a comparison, and standard errors are omitted because each placebo replication refits the weights over 11,091 control units.

```
sdid_cohort <- function(y, gg) {
  dg <- dat %>% filter(g == 0 | g == gg)
  w <- sdid_weights(dg, id = "userid", time = "weeks_join", y = y, d = "upgraded")
  c(estimate = w$estimate, n1 = w$N1, post = 15 - gg + 1)
}
sdid_tab <- lapply(setNames(outcomes6, outcomes6), function(y) {
  cells <- sapply(3:7, function(gg) sdid_cohort(y, gg))
  wts <- cells["n1", ] * cells["post", ]
  list(by_cohort = cells["estimate", ], overall = sum(cells["estimate", ] * wts) / sum(wts))
})
```

Table 8: Appendix Table C2 of the paper, in R: synthetic DiD estimates by adoption cohort (cohort 2 excluded: one pre-period) and their treated-cell-weighted combination, with the Callaway-Sant'Anna ATT for comparison.

Variable	g = 3	g = 4	g = 5	g = 6	g = 7	SDID overall	CS ATT
Exercise tracking	0.086	0.062	0.109	0.091	0.085	0.083	0.073
Food tracking	0.070	0.146	0.158	0.179	0.270	0.136	0.076
Within budget	0.055	0.076	0.095	0.177	0.221	0.097	0.068
Exercise calories	13.256	20.557	4.586	41.727	47.865	20.741	14.344
Positive weight loss	0.061	-0.025	-0.045	0.027	0.101	0.016	0.023
Weight loss amount	0.304	0.574	0.302	0.985	2.484	0.628	0.322

Synthetic DiD lands near Callaway-Sant'Anna for the exercise and health outcomes and above it for food tracking and budget adherence. It uses no covariates and builds its counterfactual by reweighting never-adopters to match each cohort's pre-period path, so it differs most where the week-1 covariates matter for selection. Later cohorts, which upgrade on rising trajectories, also show the largest cohort estimates, echoing Figure 5.

7.5 Appendix D: the TWFE coefficient tables and the week-6 exposure design

Online Tables D1 and D2 print the coefficients behind the left panels of Figures 2 and 3. They come straight from the same fits.

Online Figures D1 and D2 repeat the exposure-matched design for week-6 adopters (GAM matching, ratios 4 and 3, seed 123). Food and exercise tracking are shown; the pattern matches the week-5 figures.

Table 9: Online Tables D1-D2 of the paper: TWFE event-study coefficients for all six outcomes (standard errors in parentheses, clustered by user). Reference: never-adopters and event times at or below -4.

Event time	Exercise tracking	Food tracking	Within budget	Exercise calories	Positive weight loss	Weight loss amount
-3	0.010 (0.028)	0.092 (0.028)	0.109 (0.026)	16.635 (13.832)	0.117 (0.026)	1.376 (0.514)
-2	0.067 (0.029)	0.132 (0.035)	0.156 (0.027)	36.957 (14.092)	0.089 (0.028)	2.151 (0.588)
-1	0.063 (0.027)	0.189 (0.036)	0.175 (0.028)	38.263 (14.440)	0.071 (0.027)	2.117 (0.636)
0	0.182 (0.029)	0.293 (0.035)	0.247 (0.030)	68.134 (14.755)	0.104 (0.027)	2.344 (0.657)
1	0.186 (0.028)	0.295 (0.038)	0.277 (0.032)	72.755 (14.351)	0.096 (0.026)	2.496 (0.668)
2	0.144 (0.029)	0.298 (0.039)	0.263 (0.032)	47.552 (14.434)	0.099 (0.025)	2.778 (0.680)
3	0.138 (0.028)	0.307 (0.040)	0.235 (0.033)	52.102 (15.075)	0.098 (0.026)	2.845 (0.688)
4	0.137 (0.029)	0.288 (0.042)	0.249 (0.033)	56.464 (14.921)	0.090 (0.026)	2.924 (0.721)
5	0.111 (0.029)	0.296 (0.041)	0.257 (0.033)	50.767 (16.054)	0.084 (0.026)	2.907 (0.743)
6	0.112 (0.029)	0.279 (0.040)	0.269 (0.034)	50.321 (15.004)	0.091 (0.026)	3.026 (0.742)
7	0.121 (0.029)	0.274 (0.043)	0.253 (0.035)	46.509 (14.972)	0.078 (0.026)	3.216 (0.771)
8	0.110 (0.030)	0.269 (0.042)	0.264 (0.033)	33.232 (15.626)	0.081 (0.026)	2.991 (0.783)
9	0.086 (0.029)	0.208 (0.042)	0.221 (0.034)	31.549 (14.442)	0.081 (0.026)	2.541 (0.755)
10	0.093 (0.030)	0.234 (0.041)	0.235 (0.034)	45.032 (15.267)	0.069 (0.027)	2.296 (0.750)
11	0.071 (0.031)	0.227 (0.042)	0.229 (0.034)	45.367 (15.872)	0.065 (0.026)	2.586 (0.751)
12	0.070 (0.031)	0.178 (0.043)	0.212 (0.036)	36.082 (16.527)	0.084 (0.027)	2.479 (0.783)
13+	0.089 (0.033)	0.181 (0.044)	0.193 (0.037)	53.175 (18.649)	0.076 (0.029)	2.684 (0.880)

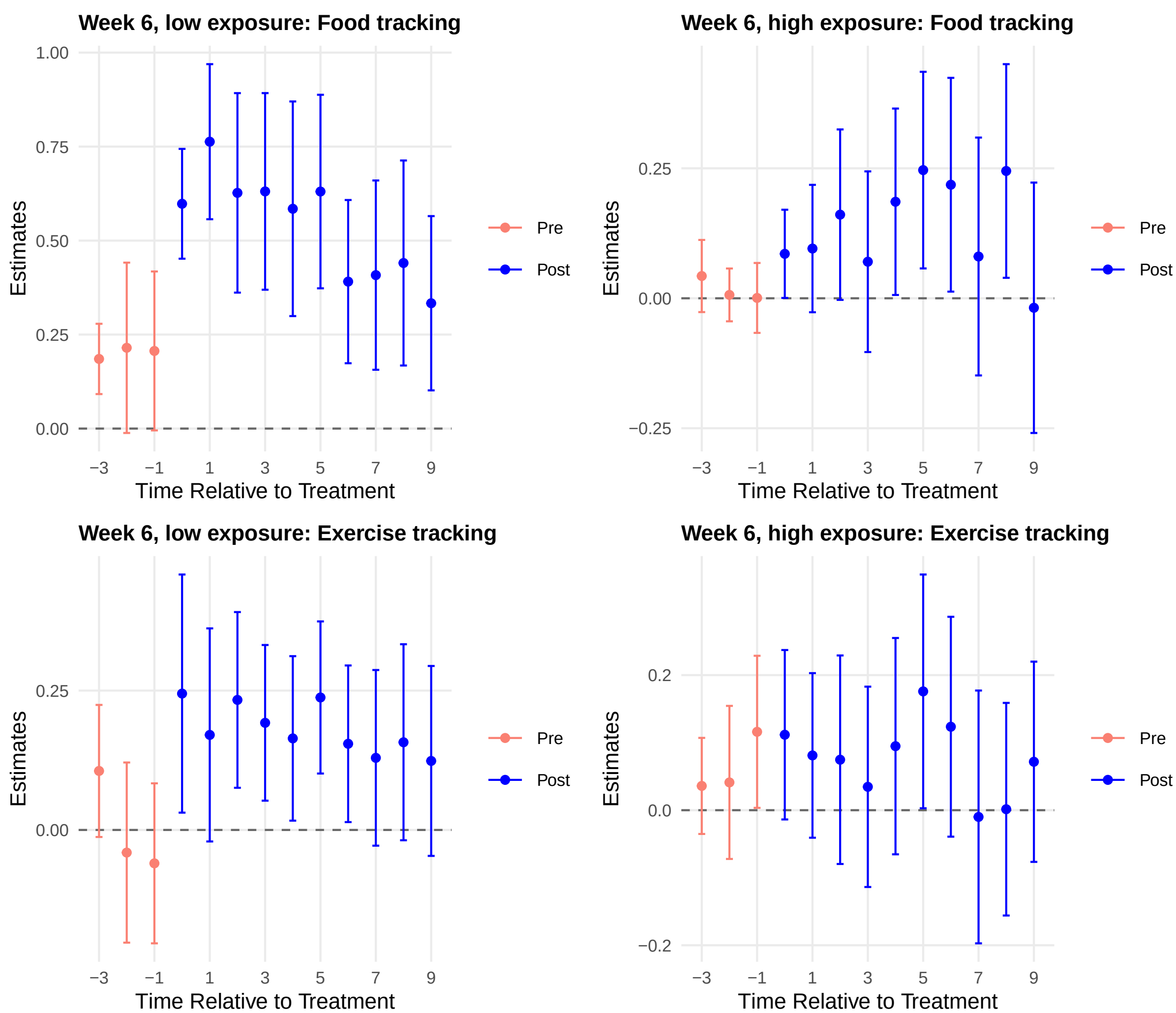


Figure 10: Online Figures D1-D2 of the paper: the exposure-matched TWFE design for week-6 adopters, food and exercise tracking. Left: low exposure; right: high exposure.

8 Extensions

Eight extensions, each answering a question the paper raises but does not settle, all on the same sample and labelled as ours.

8.1 Where the TWFE-CS gap comes from

The Goodman-Bacon decomposition writes the pooled TWFE as a weighted average of all two-by-two comparisons, and the forbidden ones carry the overstatement. `bacon_decomp()` gives every comparison's weight and estimate; `twfe_weights()` gives the de Chaisemartin-d'Haultfoeuille cell weights and their negative share.

```
bacon_food <- bacon_decomp(dat, id = "userid", time = "weeks_join", y = "track_food_avg", d = "upgraded")
bacon_food$by_type
#>                        type      weight   estimate
#> 1 earlier vs later treated 0.001423760 0.0255968
#> 2 later vs earlier treated 0.009649913 0.2403819
#> 3 treated vs never treated 0.988926327 0.1467291
tw_w <- twfe_weights(dat, id = "userid", time = "weeks_join", d = "upgraded", y = "track_food_avg")
c(share_negative = tw_w$share_negative, sum_negative = tw_w$sum_negative)
#> share_negative   sum_negative
#>              0              0
```

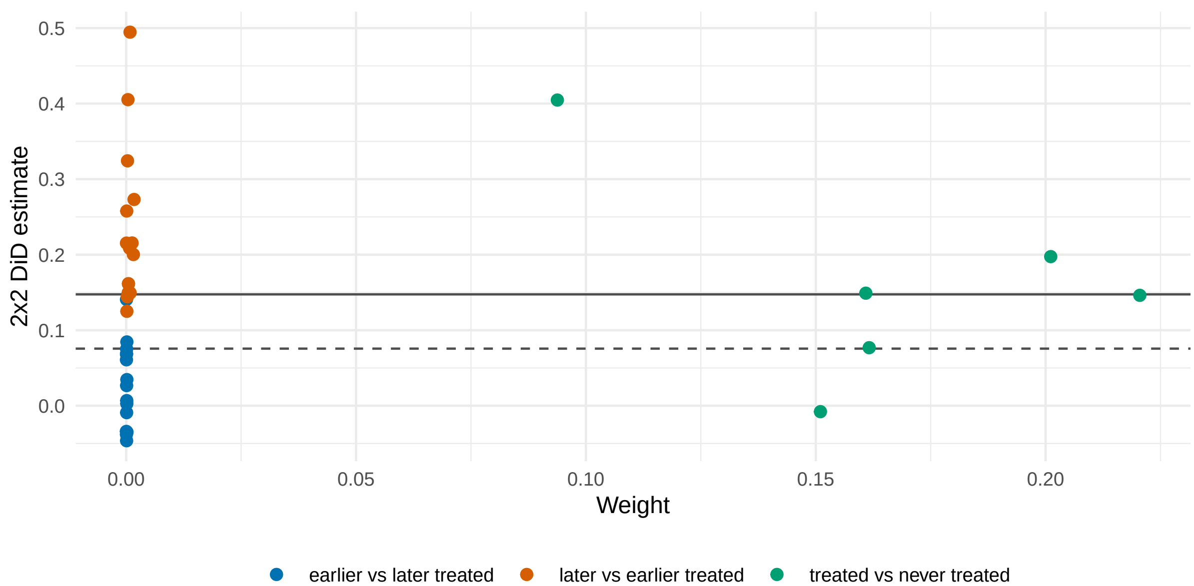


Figure 11: Extension 1: Goodman-Bacon decomposition of the pooled TWFE for food tracking. Each point is one two-by-two comparison; the horizontal line is the TWFE coefficient, the dashed line the Callaway-Sant'Anna ATT.

The decomposition acquits the usual suspects and convicts the aggregation. Negative weights are absent (0.0 percent of cells), and the comparisons that use already-treated units as controls carry about 1.1 percent of the weight. Instead, 98.9 percent of the TWFE coefficient is the static treated-versus-never comparison, which averages the whole post-adoption window without covariates. The Callaway-Sant'Anna ATT differs because it adjusts for the week-1 covariates and weights the group-time cells by cohort size over a defined window, not because forbidden comparisons dominate here.

8.2 Which cohorts carry the effect

Group and calendar aggregations, and a balanced event study, unpack the composition of Figures 2 and 3. Later cohorts (the ones with rising pre-adoption engagement in Figure 5) contribute only short horizons; the balanced aggregation holds the cohort mix fixed across event times.

```
grp <- lapply(cs_main[eng4], aggregate_att, type = "group", seed = 678)
bal <- lapply(cs_main[c("track_food_avg", "track_exercise_avg")], aggregate_att,
              type = "dynamic", min_e = -3, balance_e = 8, seed = 678)
```

Table 10: Extension 2: cohort-specific ATTs (average post-treatment effect per adoption cohort), standard errors in parentheses.

Outcome	g = 2	g = 3	g = 4	g = 5	g = 6	g = 7
Exercise tracking	0.092 (0.024)	0.071 (0.025)	0.058 (0.026)	0.062 (0.043)	0.082 (0.030)	-0.018 (0.058)
Food tracking	0.013 (0.022)	0.082 (0.031)	0.131 (0.043)	0.130 (0.056)	0.100 (0.052)	0.184 (0.082)
Within budget	0.059 (0.022)	0.051 (0.028)	0.092 (0.029)	0.053 (0.057)	0.088 (0.047)	0.109 (0.082)
Exercise calories	33.804 (13.337)	11.055 (14.198)	10.137 (15.224)	-72.029 (61.568)	34.946 (17.105)	32.378 (29.667)

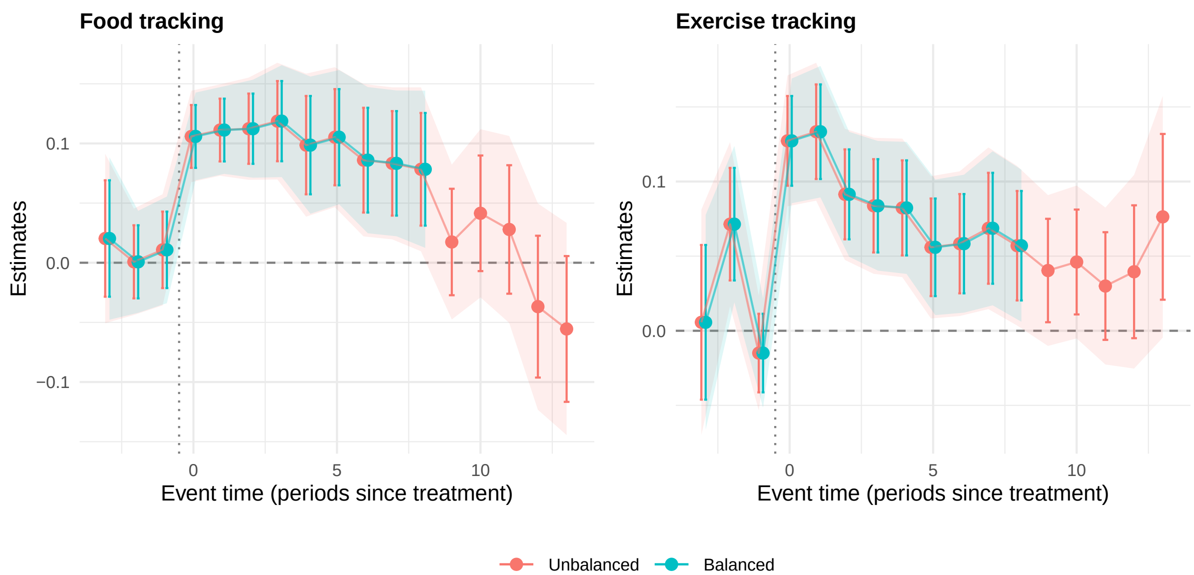


Figure 12: Extension 2: unbalanced against balanced (eight post periods per cohort) dynamic aggregation. The unbalanced curve changes cohort composition as the horizon grows; the balanced curve holds it fixed.

The attenuation is not a composition artifact. The balanced curve attenuates like the unbalanced one over the common window, so the decay in Figure 2 is within cohorts, not a changing mix of them.

8.3 How much power the pre-trend evidence has

Three pre-period estimates cannot rule out much, and Roth's (2022) calculation says how much. For each outcome, `pretrend_power()` finds the linear violation that the pre-test would detect with 50 and 80 percent power and the bias such an undetected trend would leave in the week-0 effect.

```
pow <- lapply(dyn_main[c("track_food_avg", "track_exercise_avg")], function(d) {
  p0 <- pretrend_power(d, seed = 1)
  s50 <- p0$detectable$slope[p0$detectable$target_power == 0.5]
  p1 <- pretrend_power(d, slope = s50, seed = 1)
  list(detectable = p0$detectable, bias0 = p1$slopes[[1]]$bias)
})
```

A violation detected only half the time would move the week-0 effect by a small fraction of its size. That is why Figure 7's conclusions survive moderate $\bar{M}$: the effects are an order of magnitude larger than the trends the pre-periods could hide.

Table 11: Extension 3: power of the pre-trend evidence (Roth 2022). The detectable slope is the linear violation per week the three pre-period coefficients would flag with the stated power; the last columns give the week-0 bias such a trend would leave, unconditionally and conditional on the pre-test passing.

Outcome	Slope detected, 50% power	Slope detected, 80% power	Week-0 effect	Bias at 50%-power slope	Conditional bias (test passed)
Food tracking	0.0193	0.0283	0.106	0.0193	0.0168
Exercise tracking	0.0206	0.0300	0.127	0.0206	0.0204

8.4 Breakdown values by outcome and horizon

Figure 7 covers event time 0; the honest question extends across horizons and to the smoothness family. For each engagement outcome the table reports the relative-magnitudes breakdown value (the smallest $\bar{M}$ at which zero enters the interval) at event time 0 and for the average of event times 0 to 3, and the smoothness breakdown M at event time 0.

```
breakdown_rm <- function(dyn, l_vec) {
  hi <- honestdid_inputs(dyn)
  rob <- HonestDiD::createSensitivityResults_relativeMagnitudes(betahat = hi$betahat, sigma = hi$sigma,
    numPrePeriods = hi$numPrePeriods, numPostPeriods = hi$numPostPeriods,
    Mbarvec = seq(0.2, 3, by = 0.1), l_vec = l_vec, gridPoints = 100)
  cov0 <- rob$lb <= 0 & rob$ub >= 0
  if (any(cov0)) min(rob$Mbar[cov0]) else Inf
}
breakdown_sm <- function(dyn, l_vec, m_max) {
  hi <- honestdid_inputs(dyn)
  rob <- HonestDiD::createSensitivityResults(betahat = hi$betahat, sigma = hi$sigma,
    numPrePeriods = hi$numPrePeriods, numPostPeriods = hi$numPostPeriods,
    Mvec = seq(0, m_max, length.out = 11), l_vec = l_vec)
  cov0 <- rob$lb <= 0 & rob$ub >= 0
  if (any(cov0)) min(rob$M[cov0]) else Inf
}
honest_ext <- lapply(setNames(eng4, eng4), function(y) {
  d <- dyn_uni[[y]]
  np <- honestdid_inputs(d)$numPostPeriods
  l0 <- HonestDiD::basisVector(1, np)
  lavg <- rep(1 / np, np)
  m_max <- if (y == "exercisecalories_avg") 25 else if (y == "track_exercise_avg") 0.15 else 0.06
  c(rm0 = breakdown_rm(d, l0), rm_avg = breakdown_rm(d, lavg), sm0 = breakdown_sm(d, l0, m_max))
})
```

Table 12: Extension 4: breakdown values. RM columns: the smallest Mbar (violations up to Mbar times the largest pre-period deviation) at which zero enters the confidence set. Smoothness column: the smallest per-week slope change M with the same property.

Outcome	RM breakdown, e = 0	RM breakdown, avg e = 0-3	Smoothness breakdown, e = 0
Exercise tracking	1.5	0.5	0.105
Food tracking	1.8	0.8	0.060
Within budget	0.8	0.4	0.030
Exercise calories	0.8	0.2	0.000

Averaging over the first month is less robust than the impact effect, because later horizons let violations accumulate. A smoothness breakdown of zero for exercise calories says a continued linear pre-trend alone removes its significance, matching its weaker relative-magnitudes robustness. This sharpens the paper's reading: the credible finding is the short-lived lift, not its persistence.

8.5 The matched sample with the main estimator

Figure 8 switches estimators along with samples; holding the estimator fixed isolates what the matching does. `att_gt()` runs on the matched panel of Section 5.4, and the frame overlays it on the matched-sample TWFE.

```
cs_m <- lapply(setNames(c("track_food_avg", "track_exercise_avg"), c("track_food_avg", "track_exercise_avg")),
  function(y) {
    a <- att_gt(dat8, id = "userid", time = "weeks_join", group = "g", y = y, x = xv,
                method = "dr", control_group = "notyet", n_boot = 499, seed = 678)
    aggregate_att(a, type = "dynamic", min_e = -3, na.rm = TRUE, seed = 678)
  })
```

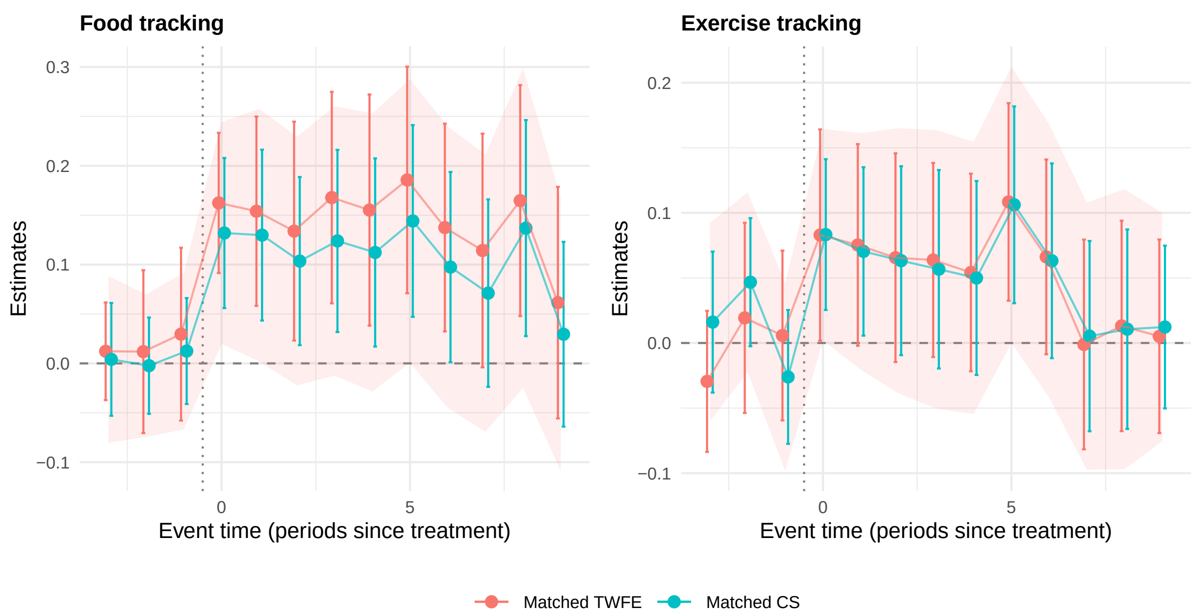


Figure 13: Extension 5: the matched sample of Figure 8 estimated with Callaway-Sant'Anna next to the matched-sample TWFE.

The two agree on the matched sample, so the robustness claim does not depend on switching estimators. The lift and the attenuation are in the design, not the method.

8.6 Anticipation

Figure 5's rising pre-adoption activity could be anticipation rather than selection. `att_gt(anticipation = 1)` treats the week before adoption as already affected and moves the base period one week back.

```
ant <- lapply(setNames(c("track_food_avg", "track_exercise_avg"), c("track_food_avg", "track_exercise_avg")),
  function(y) {
    a <- att_gt(dat, id = "userid", time = "weeks_join", group = "g", y = y, x = xv,
                method = "dr", control_group = "notyet", anticipation = 1, n_boot = 499, seed = 678)
    aggregate_att(a, type = "dynamic", min_e = -3, na.rm = TRUE, seed = 678)$by
  })
```

Exercise tracking is stable; the food-tracking impact effect grows when the decision week counts as treated. With one week of anticipation the food effect at adoption rises from 0.106 to 0.140, which says part of the food response in this draw begins in the week the upgrade happens rather than strictly after it. The baseline week -1 coefficient is small and insignificant, so the lift-then-decay reading survives either dating; the exercise is a reminder that the anticipation window is a modelling choice worth reporting.

8.7 Exposure as a gradient

The median split of Table 5 is the coarsest version of the hedonic decline test; terciles sharpen it. Adopters are split into thirds by pre-adoption tracking days, and the week-0 and week-1 effects are

Table 13: Extension 6: allowing one week of anticipation. The base period moves from g-1 to g-2; a large change in the week-0 effect would signal that the g-1 outcome is already treated.

Outcome	Event time	Baseline	Anticipation = 1
Food tracking	-1	0.011 (0.016)	0.011 (0.015)
	0	0.106 (0.013)	0.140 (0.019)
	1	0.111 (0.013)	0.145 (0.021)
	2	0.112 (0.015)	0.144 (0.019)
Exercise tracking	-1	-0.015 (0.013)	-0.015 (0.015)
	0	0.127 (0.015)	0.116 (0.021)
	1	0.133 (0.016)	0.109 (0.020)
	2	0.091 (0.015)	0.051 (0.019)

estimated per tercile.

```
adopters <- users %>% filter(week_upgrade >= 2, week_upgrade <= 7)
terc <- quantile(adopters$total_track_before, c(1/3, 2/3))
terc_of <- function(v) cut(v, c(-Inf, terc, Inf), labels = c("Low", "Middle", "High"))
terc_run <- function(lev) {
  ids <- adopters$userid[terc_of(adopters$total_track_before) == lev]
  dd <- dat %>% filter(userid %in% c(ids, never_ids))
  lapply(setNames(c("track_food_avg", "track_exercise_avg"), c("track_food_avg", "track_exercise_avg")),
    function(y) {
      a <- att_gt(dd, id = "userid", time = "weeks_join", group = "g", y = y, x = xv,
                  method = "dr", control_group = "notyet", n_boot = 499, seed = 1234)
      aggregate_att(a, type = "dynamic", min_e = -3, na.rm = TRUE, seed = 1234)$by
    })
}
terc_res <- lapply(setNames(c("Low", "Middle", "High"), c("Low", "Middle", "High")), terc_run)
```

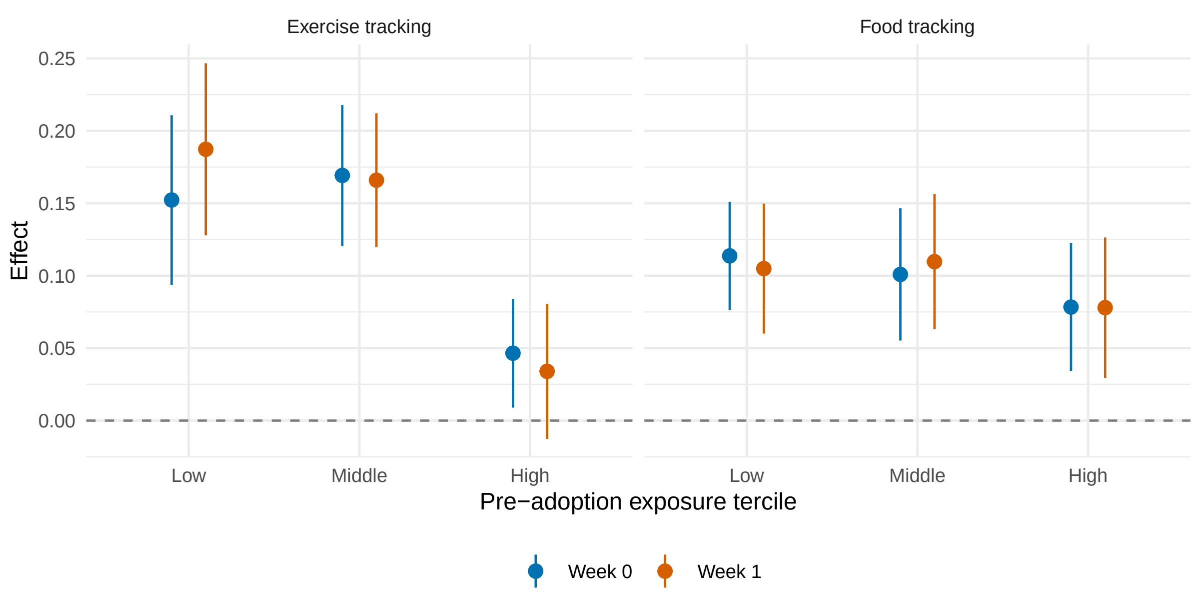


Figure 14: Extension 7: week-0 and week-1 effects by tercile of pre-adoption exposure, with 95 percent intervals. Hedonic decline predicts the monotone decline in the lift; sunk costs predict a flat profile.

The high-exposure tercile sits well below the other two; low and middle are indistinguishable. The profile is a step down at high exposure rather than a straight line, which still contradicts the flat profile sunk costs predict: users who had already exhausted the shared features respond least. The novelty margin appears to bind only once exposure is high.

8.8 An estimator horse race

The lecture's comparison of heterogeneity-robust estimators, run on this application. Callaway-Sant'Anna (covariates, not-yet-treated), Sun-Abraham (interaction-weighted, `fixest::sunab()`), the imputation estimator (`did_imputation()`), and the binned TWFE, all for food tracking.

```
dat$g_sa <- ifelse(dat$g == 0, 10000L, dat$g)
sa <- feols(track_food_avg ~ sunab(g_sa, weeks_join) | userid + weeks_join, data = dat, cluster = ~userid)
imp <- did_imputation(dat, id = "userid", time = "weeks_join", group = "g", y = "track_food_avg",
                      pre_window = 3)
race <- event_study_frame(`Callaway-Sant'Anna` = dyn_main[["track_food_avg"]],
                          `Sun-Abraham` = sa, Imputation = imp,
                          `TWFE (binned)` = tw_main[["track_food_avg"]])
```

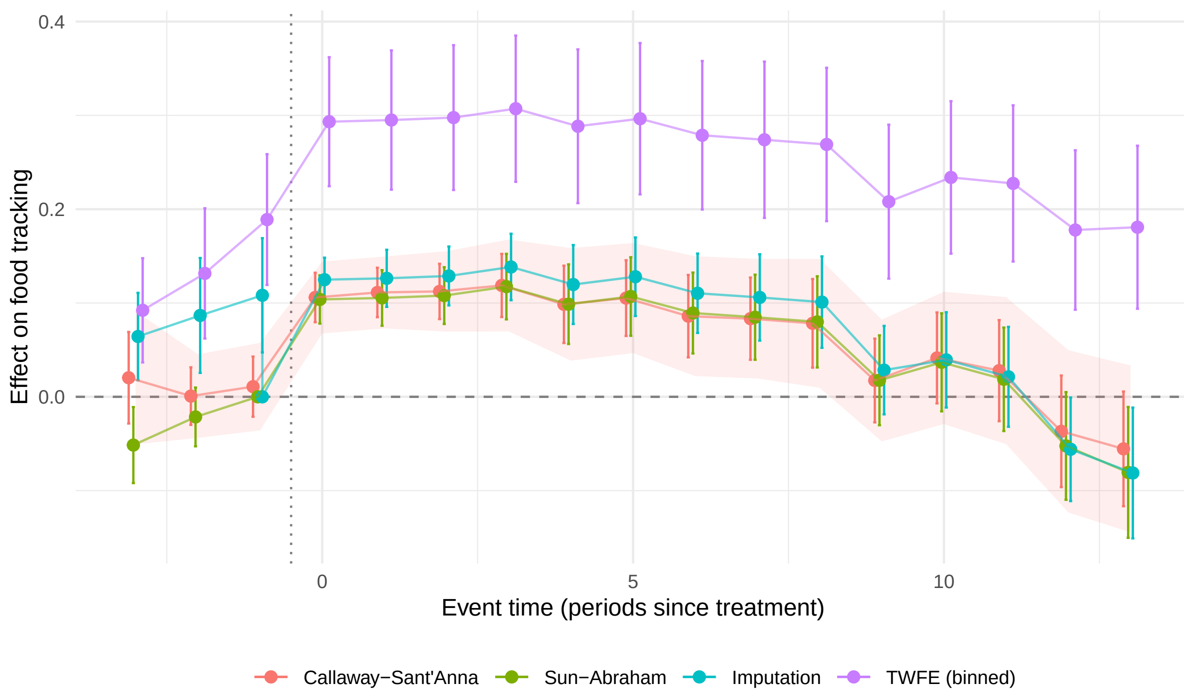


Figure 15: Extension 8: four estimators of the food-tracking event study. The binned TWFE sits far above the three heterogeneity-robust estimators at every horizon.

Table 14: Extension 8: the food-tracking event study at four horizons across estimators (standard errors in parentheses).

Estimator	e = 0	e = 1	e = 5	e = 10
Callaway-Sant'Anna	0.106 (0.013)	0.111 (0.013)	0.105 (0.021)	0.041 (0.025)
Sun-Abraham	0.104 (0.013)	0.105 (0.015)	0.107 (0.021)	0.037 (0.027)
Imputation	0.125 (0.012)	0.126 (0.016)	0.128 (0.021)	0.039 (0.026)
TWFE (binned)	0.293 (0.035)	0.295 (0.038)	0.296 (0.041)	0.234 (0.041)

Callaway-Sant'Anna and Sun-Abraham agree to the third decimal, the imputation estimator sits slightly above them, and the binned TWFE is far above all three at every horizon. The TWFE gap is the reference group: its coefficients compare against never-adopters and distant pre-periods jointly, so cohort-level differences load onto the event dummies. The imputation estimator fits the untreated model on all pre-periods, which buys precision at the price of pre-period parallel trends; its closeness to Callaway-Sant'Anna says that assumption is not doing damage here. Synthetic DiD, the remaining estimator, is in Section 7.4.

9 What the replication says

The workflow reproduces end to end on the package. On the simulated data our `att_gt()` matches the `did` package to machine precision, the TWFE helper matches the authors' dummy regressions exactly, and every table and figure of the paper and its appendix is rebuilt in the published layout with `att_gt()`, `aggregate_att()`, `event_study_frame()`, `plot_event_study()`, and `honestdid_inputs()`, plus MatchIt as the matching engine and HonestDiD, DIDmultiplegtDYN, and `sdid_weights()` for the sensitivity and alternative estimators.

The paper's lessons survive the added cuts, with two honest qualifications from the simulated draw. The TWFE-versus-CS gap is the static, covariate-free treated-versus-never comparison, not negative weights or forbidden controls; the attenuation is within cohorts, not composition, though this draw damps the food-tracking decay; the exposure effect is a step down at high exposure rather than a monotone gradient; the matched sample gives the same answer under the main estimator; one week of anticipation leaves exercise unchanged and grows the food impact effect, a dating choice worth reporting; and Sun-Abraham and the imputation estimator agree with Callaway-Sant'Anna where their assumptions overlap.

What to remember for one's own staggered designs. Drop cohorts without pre-periods; state the comparison group; put covariates in both nuisances; aggregate deliberately (dynamic, group, balanced); report simultaneous bands; quantify how much the pre-periods can actually rule out before calling trends parallel; and treat TWFE as a diagnostic, not an estimate.

10 References

- Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., and Sant'Anna, P. H. C. 2026. "Difference-in-Differences Designs: A Practitioner's Guide." Working paper.
- Callaway, B., and Sant'Anna, P. H. C. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*.
- de Chaisemartin, C., and d'Haultfoeuille, X. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review*.
- Goodman-Bacon, A. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics*.
- Jiang, Y., Uetake, K., and Yang, N. 2026. "Does Premium Version Adoption in mHealth Improve User Engagement and Health-Related Outcomes?" *Marketing Science*, Articles in Advance. Replication package: INFORMS.
- Rambachan, A., and Roth, J. 2023. "A More Credible Approach to Parallel Trends." *Review of Economic Studies*.
- Roth, J. 2022. "Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends." *AER: Insights*.
- Sun, L., and Abraham, S. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics*.